

# Recursive Intelligence-Field Theory (RIFT): Observational Constraints from BAO, CMB, and Large-Scale Structure

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## Abstract

Current cosmological theory relies on an uneasy union between General Relativity (GR) and the  $\Lambda$ CDM model, requiring dark matter, dark energy, and inflation—none of which have been directly confirmed in laboratory physics. This paper presents the *Recursive Intelligence-Field Theory* (RIFT), a covariant scalar–tensor field theory in which spacetime curvature is dynamically coupled to a scalar field  $\Psi$  that recursively encodes the history of past curvature. The theory is derived from a single covariant action

$$S = \int d^4x \sqrt{-g} \left[ \left( \frac{1}{16\pi G} + \Lambda(\Psi) \right) R + \frac{1}{2}(\nabla\Psi)^2 - \frac{1}{2}m(\Psi)^2\Psi^2 - V(\Psi) \right] + S_{\text{matter}},$$

from which the gravitational and scalar field equations follow by standard variation—no additional postulates are required. The non-minimal coupling  $\Lambda(\Psi)R$  produces curvature memory: geometry at any event retains a causal integral over past curvature, with causality enforced by the retarded Green’s function. Imposing GR-recovery conditions ( $\Lambda(0) = 0$ ) and numerical verification of the full FLRW system (Sims 82–86) confirm the theory is self-consistent and reduces to GR in the  $\Psi \rightarrow 0$  limit.

We subject RIFT to twelve independent observational tests (Sims 87–98). The key results are: (i) BAO full-covariance refit (BOSS+eBOSS+Ly $\alpha$ ):  $\chi^2/\text{dof} = 1.22$ ,  $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = 0$ ; (ii) CMB TT power spectrum via full CLASS Boltzmann solver: RMS deviation = 2.93% ( $\ell = 2\text{--}1500$ ),  $G_{\text{eff}}/G = 1 - 16$  ppm at the best-fit coupling; (iii) joint CMB+BAO parameter fit:  $\Delta\chi^2 = 0$ , best-fit  $H_0 = 67.59$  km/s/Mpc,  $\Omega_m = 0.312$ ,  $\Lambda_0 \approx 0.013$  (numerically unconstrained;  $\chi^2$  flat across  $[0, 0.1]$ ); (iv) non-linear structure growth:  $|\Delta\sigma_8| < 0.007\%$  at  $\Lambda_0 = 0.003$ , RIFT provides no relief for the Planck/KiDS  $S_8$  tension; (v) Bayesian model comparison:  $\ln B = -0.71$  (Inconclusive; RIFT not excluded); (vi) DESI Year 1 BAO refit:  $\chi^2/\text{dof} = 1.36$ ,  $\Delta\chi^2 = 0$ , joint CMB+DESI best-fit consistent with BOSS-based result at  $< 0.2\sigma$ ; (vii) CMB polarization EE/TE via CLASS: RMS deviations  $< 0.2\%$  (EE) and  $< 0.5\%$  (TE) at joint best-fit; (viii) RSD/ $f\sigma_8$  growth rate (9 surveys,  $z = 0.07\text{--}1.48$ ):  $\chi^2/\text{dof} = 0.86$ ,  $\Delta\chi^2 = -0.14 \approx 0$ ; (ix) official Planck 2018 plikHM TTTEEE likelihood (Sim 97, via `clipy`):  $\Delta(-2\ln\mathcal{L}) = +7.0$  at the Sim 90 parameter point ( $G_{\text{eff}}/G = 1 - 42$  ppm is negligible; driven by parameter offset); (x) joint plikHM + BAO re-optimisation (Sim 98):  $\Delta(-2\ln\mathcal{L}) =$

$-0.013 \approx 0$  at the CMB-optimised best-fit ( $H_0 = 67.30$ ,  $\Omega_m = 0.317$ ,  $\Lambda_0 = 0.005$ ,  $G_{\text{eff}}/G = 1 - 24 \text{ ppm}$ ); the Sim 97 tension is fully resolved; (xi) galaxy rotation curves, static (Sim 99): the locked action at  $\Lambda_0 = 0.003$  cannot replace dark matter—the Yukawa halo profile is exponentially steeper than the  $1/r^2$  isothermal needed for flat curves; (xii) galaxy rotation curves, time-domain PDE (Sim 100): galactic curvature coupling  $2\Lambda_0|R_{\text{gal}}| \approx 5 \times 10^{-9} \text{ kpc}^{-2}$  is  $5 \times 10^7$  times weaker than  $m_0^2$ ; curvature condensation at galactic densities is ruled out;  $\lambda < 0$  self-coupling causes tachyonic blow-up; (xiii) BAO sound horizon from first principles (Sim 101):  $|\Delta r_d/r_d| < 21 \text{ ppm}$  across all allowed  $\Lambda_0$ ; RIFT and  $\Lambda\text{CDM}$  are degenerate in  $r_d$ ; (xiv) one-loop UV structure, partial (Sim 102): quartic UV divergence confirmed; memory damping regulates to  $\Sigma(0) = \Lambda_0^2 k_m^4 / (64\pi^2)$ ; naturalness requires  $\tau_m > 0.11 \text{ Mpc}$ ; (xv) full one-loop UV structure, all diagrams (Sim 104): graviton wfn renorm protected by diffeomorphism Ward identity  $\Pi_{hh}(0) = 0$ ; vertex correction  $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$  finite; two-loop  $\Sigma_D/m_0^2 = 3.4 \times 10^{-6}$  (perturbative hierarchy stable); UV-finiteness confirmed at full one-loop level in the resummed theory; (xvi) RG flow of  $\Lambda_0$  (Sim 105): beta function negative (asymptotically free);  $\Lambda_0 = 0$  is a UV fixed point (theory reduces to GR+scalar at high energies); analytic formula  $1/\Lambda_0^2 = 1/\Lambda_{0,\text{obs}}^2 + (k_m^2 - k_{m,\text{nat}}^2)/(16\pi^2 m_0^2)$  validated; no IR Landau pole; IR running  $< 2.5\%$ ;  $\Lambda_0 \approx 0.003$  is an IR attractor, not fine-tuned; (xvii) two-loop graviton sector (Sim 106): Ward identity  $\Pi_{hh}(0) = 0$  exact at all loop orders; resummed Psi propagator regulates all Psi-bubble insertions;  $\delta Z_h \sim 10^{-5}$  negligible; last UV caveat closed; (xviii) Vainshtein screening (Sim 103): all scanned  $(\alpha_V, M_V, \beta)$  combinations fail to source an isothermal dark matter halo for NGC 3198; field gradient  $|\delta\Psi'| \sim 10^{-8} \text{ kpc}^{-1}$  is  $10^6\times$  below the nonlinear threshold; energy density deficit is  $2.7 \times 10^6\times$ ; all kinetic modifications of the locked action are exhausted. In all cosmological tests (Sims 87–98), RIFT at the joint best-fit is statistically indistinguishable from  $\Lambda\text{CDM}$  at current precision. The recursive field coupling  $\Lambda_0$  is constrained to  $\Lambda_0 < 0.095$  (95% CL) and is cosmologically inert at current observational precision. Galactic-scale predictions require new physics beyond the locked action: a first-principles coupling between  $\Psi$  and baryonic matter. Falsifiable predictions are given for future surveys (DESI Y5, Euclid, CMB-S4).

## Contents

|          |                                                           |          |
|----------|-----------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                       | <b>4</b> |
| <b>2</b> | <b>Recursive Foundations</b>                              | <b>5</b> |
| <b>3</b> | <b>Lagrangian Formulation and Field Equations</b>         | <b>5</b> |
| 3.1      | The RIFT Action . . . . .                                 | 5        |
| 3.2      | GR Recovery . . . . .                                     | 6        |
| 3.3      | Scalar Field Equation . . . . .                           | 6        |
| 3.4      | Curvature Memory and the Retarded Solution . . . . .      | 6        |
| 3.5      | Gravitational Field Equation . . . . .                    | 6        |
| 3.6      | FLRW Reduction and Modified Friedmann Equations . . . . . | 7        |
| <b>4</b> | <b>Recursive Field Equations (Scalar Attractor)</b>       | <b>7</b> |
| <b>5</b> | <b>Canonical Quantization</b>                             | <b>8</b> |
| 5.1      | Scalar Field Quantization . . . . .                       | 8        |

|           |                                                                    |           |
|-----------|--------------------------------------------------------------------|-----------|
| 5.2       | Tensor Field Quantization (Recursive Graviton)                     | 8         |
| 5.3       | Recursive Memory Decoherence and UV Finiteness                     | 8         |
| <b>6</b>  | <b>CMB Power Spectrum from Recursive Fields</b>                    | <b>9</b>  |
| 6.1       | RIFT Primordial Power Spectrum                                     | 9         |
| 6.2       | BAO from Recursive Echo Shells                                     | 9         |
| <b>7</b>  | <b>Observational Consistency Tests</b>                             | <b>9</b>  |
| 7.1       | BAO Full-Covariance Refit (Sim 87)                                 | 10        |
| 7.2       | CMB Full CLASS Injection (Sim 88)                                  | 10        |
| 7.3       | Planck TT Likelihood (Sim 89)                                      | 11        |
| 7.4       | Joint CMB+BAO Parameter Fit (Sim 90)                               | 11        |
| 7.5       | Non-linear Structure Growth and $\sigma_8$ (Sim 92)                | 12        |
| 7.6       | DESI Year 1 BAO Refit (Sim 94)                                     | 13        |
| 7.7       | CMB Polarization EE/TE (Sim 95)                                    | 14        |
| 7.8       | RSD Growth Rate $f\sigma_8$ (Sim 96)                               | 15        |
| 7.9       | Definitive Planck 2018 plikHM TTTEEE Likelihood (Sim 97)           | 16        |
| 7.10      | Joint plikHM + BAO Parameter Re-fit (Sim 98)                       | 17        |
| <b>8</b>  | <b>Recursive Graviton Emission</b>                                 | <b>18</b> |
| <b>9</b>  | <b>Falsifiability Predictions</b>                                  | <b>18</b> |
| 9.1       | Galaxy Rotation Curves                                             | 18        |
| 9.2       | Void Lensing (DES, Euclid)                                         | 19        |
| 9.3       | Bullet Cluster Lensing Offset                                      | 19        |
| 9.4       | CMB Low- $\ell$ Anomalies                                          | 19        |
| <b>10</b> | <b>Solar System Viability: Effective Matter Coupling (Sim 107)</b> | <b>19</b> |
| 10.1      | Motivation                                                         | 19        |
| 10.2      | Effective Coupling                                                 | 19        |
| 10.3      | Results                                                            | 20        |
| <b>11</b> | <b>Gravitational Wave Speed, GW170817, and BBN (Sim 108)</b>       | <b>20</b> |
| 11.1      | GW speed from the tensor perturbation equation                     | 20        |
| 11.2      | Results                                                            | 20        |
| <b>12</b> | <b>Conclusion</b>                                                  | <b>20</b> |
| <b>A</b>  | <b>Parameter Tables and Definitions</b>                            | <b>23</b> |
| A.1       | Core Fields and Coupling Terms                                     | 23        |
| A.2       | Best-Fit Cosmological Parameters (Sim 87–90)                       | 23        |
| A.3       | Key Limiting Cases                                                 | 23        |
| <b>B</b>  | <b>Recursive Coupling to the Standard Model</b>                    | <b>24</b> |

# 1 Introduction

General Relativity (GR), while highly predictive at solar and astrophysical scales, requires three independent postulates—dark matter, dark energy ( $\Lambda$ ), and inflation—at the galactic and cosmological scales where its predictions break down. None of these entities have been detected in controlled laboratory settings; each is introduced to match a specific class of observations rather than derived from a deeper principle.

A deeper reformulation is needed: one that internalizes memory, feedback, and dynamical curvature saturation as fundamental features of spacetime itself.

This motivates the *Recursive Intelligence-Field Theory* (RIFT): a field-theoretic framework in which the geometry of spacetime is governed not solely by local energy-momentum, but by recursive memory—a saturation of curvature history encoded in a scalar field  $\Psi$ . The term “intelligence” is not anthropomorphic; it reflects the field’s recursive ability to adapt, stabilize, and store geometric information, akin to attractor dynamics in complex systems.

RIFT introduces  $\Psi$  with a non-minimal curvature coupling  $\Lambda(\Psi)R$ , a field-dependent mass  $m(\Psi)$ , and a potential  $V(\Psi)$  encoding attractor dynamics. The resulting theory unifies gravitation, entropy, and quantized information flow through a single field structure. The theory reproduces the Einstein field equations in the limit of vanishing field memory, while generating emergent gravitational behavior—flat rotation curves, modified expansion history, altered CMB acoustic peaks—in high-curvature or memory-rich regimes.

This paper develops the full Lagrangian formalism, derives the Euler–Lagrange and gravitational field equations, quantizes the scalar and tensor components, and tests the cosmological predictions against precision data.

## Results established in this paper (Sims 82–98):

- Modified CMB power spectrum via full CLASS Boltzmann solver:  $\text{RMS}(\Delta C_\ell/C_\ell) = 2.93\%$  (Sim 88);
- BAO fit with full  $12 \times 12$  covariance:  $\chi^2/\text{dof} = 1.22$  (Sim 87);
- Joint CMB+BAO consistency:  $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = 0$  (Sim 90); confirmed with real Planck 2018 plikHM TTTEEE likelihood:  $\Delta(-2\ln\mathcal{L}) = -0.013$  (Sim 98);
- Non-linear structure growth:  $|\Delta\sigma_8| < 0.007\%$  at  $\Lambda_0 = 0.003$  (Sim 92);
- Bayesian model comparison:  $\ln B = -0.71$  (Inconclusive; Sim 93).

## Predictions for future simulation (not yet demonstrated):

- Galaxy rotation curves without dark matter: RIFT predicts flat curves from the radial gradient  $\nabla_r\Psi$ . Sim 99 tested this explicitly: the locked action at  $\Lambda_0 = 0.003$  does not replace dark matter via  $T^\Psi$  alone; the Yukawa profile is exponentially steeper than the  $1/r^2$  isothermal sphere needed for curve flattening (Section ??). Galactic-scale predictions await derivation of a screening or condensation mechanism from the locked action.
- UV finiteness of the quantized theory: the recursive propagator is conjectured to have improved UV behaviour, but a complete loop calculation has not been performed (Section 5).
- Recursive graviton emission thresholds (Section 8).

## 2 Recursive Foundations

Contemporary physics treats spacetime as a geometric stage on which fields and particles evolve. In GR, curvature arises instantaneously from energy and momentum, with no intrinsic memory of prior configurations. Likewise, quantum field theory (QFT) operates on a fixed background, with no self-modifying or feedback-sensitive geometry. Yet nearly all complex systems in nature—including life, computation, and cognition—are recursive: they possess internal memory, feedback, and attractor stabilization that guide evolution over time.

The Recursive Intelligence-Field Theory introduces a radical shift: spacetime itself is a recursive, informationally active system. The field  $\Psi$  is not merely an auxiliary scalar but the core carrier of memory, curvature history, and geometric coherence. *Recursion* here refers to the property that the field’s evolution depends not only on its present state and local derivatives, but also on accumulated geometric information—what the field *remembers* of curvature dynamics.

By treating  $\Psi$  as a memory-encoding agent, RIFT reframes gravitation as emergent from recursion, not from mass-energy alone. The geometry of spacetime becomes a dynamic ledger of curvature history, modulated by recursive potentials and stabilized by attractor feedback. This is the mechanism by which observed gravitational phenomena—flat galaxy rotation curves, void lensing anomalies, and early-universe coherence—are predicted to arise without invoking invisible matter or finely tuned scalar inflation, though the galactic-scale predictions await dedicated simulation.

RIFT is not a rejection of GR or QFT, but their recursive extension. It maintains covariance, admits canonical quantization, and yields Einstein’s equations as a limit. But it posits that the fundamental currency of the universe is not mass or energy, but recursive memory: a field-driven intelligence embedded in spacetime itself.

## 3 Lagrangian Formulation and Field Equations

### 3.1 The RIFT Action

We work throughout in metric signature  $(-, +, +, +)$  with  $\square \equiv \nabla^\mu \nabla_\mu$  and units in which  $c = 1$ . The complete covariant action of RIFT is

$$S = \int d^4x \sqrt{-g} \left[ \left( \frac{1}{16\pi G} + \Lambda(\Psi) \right) R + \frac{1}{2} (\nabla_\mu \Psi)(\nabla^\mu \Psi) - \frac{1}{2} m(\Psi)^2 \Psi^2 - V(\Psi) \right] + S_{\text{matter}}, \quad (1)$$

where  $\Psi(x^\mu)$  is the *recursive intelligence field*, a real scalar that encodes the local history of spacetime curvature. The functions  $\Lambda(\Psi)$ ,  $m(\Psi)$ , and  $V(\Psi)$  are smooth and satisfy the GR recovery conditions stated in Section 3.2.

The physical role of each term:

- $R/(16\pi G)$ : Einstein-Hilbert term. Standard gravity in the absence of field memory.
- $\Lambda(\Psi)R$ : non-minimal curvature coupling. Modifies the effective gravitational constant. Requires  $\Lambda(\Psi) > -1/(16\pi G)$  so that  $G_{\text{eff}} > 0$ .
- $\frac{1}{2}(\nabla_\mu \Psi)(\nabla^\mu \Psi)$ : canonical kinetic term; causal propagation.

- $-\frac{1}{2}m(\Psi)^2\Psi^2$ : field-dependent mass. The minus sign gives stable dispersion  $\omega^2 = \mathbf{k}^2 + m_{\text{eff}}^2 > 0$  (verified Sim 83). The v1 draft had the wrong sign ( $+\frac{1}{2}m^2\Psi^2$ ), which would give imaginary frequency.
- $-V(\Psi)$ : recursive potential. Attractor dynamics and nonlinear saturation.

### 3.2 GR Recovery

We require  $\Lambda(0) = 0$ ,  $G_{\text{eff}}(\Psi) = G/(1 + 16\pi G\Lambda(\Psi)) > 0$ , and  $V(0) = 0$ . Under these conditions:

$$\Lambda \rightarrow 0, \quad T_{\mu\nu}^{(\Psi)} \rightarrow 0, \quad \nabla_\mu \nabla_\nu \Lambda \rightarrow 0 \implies G_{\mu\nu} = 8\pi G T_{\mu\nu}^{(\text{m})}. \quad (2)$$

Coupling forms satisfying this include  $\Lambda(\Psi) = \Lambda_0\Psi^2$ ,  $\Lambda_0\Psi^2/(1 + \Psi^2)$ ,  $\Lambda_0 \tanh(\Psi^2)$ , and  $\Lambda_0(1 - e^{-\beta\Psi^2})$ . All were numerically verified (Sim 84); the form  $\Lambda_0 e^{-\beta\Psi^2}$  (which gives  $\Lambda(0) = \Lambda_0 \neq 0$ ) was correctly rejected.

### 3.3 Scalar Field Equation

Varying (1) with respect to  $\Psi$  gives

$$\square\Psi + m_{\text{eff}}^2(\Psi)\Psi = \Lambda'(\Psi)R + V'(\Psi), \quad (3)$$

where  $m_{\text{eff}}^2(\Psi) \equiv m(\Psi)[m(\Psi) + \Psi m'(\Psi)]$ . In flat space,  $(\square + m_{\text{eff}}^2)\Psi = 0$  gives  $\omega^2 = \mathbf{k}^2 + m_{\text{eff}}^2 > 0$ , confirming stability.

### 3.4 Curvature Memory and the Retarded Solution

Equation (3) is a wave equation sourced by spacetime curvature  $R$ . Imposing retarded boundary conditions selects the unique causal solution

$$\boxed{\Psi(x) = \int G_R(x, x') [\Lambda'(\Psi(x')) R(x') + V'(\Psi(x'))] d^4x',} \quad (4)$$

where the retarded Green's function satisfies  $(\square_x + m_{\text{eff}}^2)G_R(x, x') = \delta^{(4)}(x - x')/\sqrt{-g}$  and  $G_R(x, x') = 0$  for  $x^0 < x'^0$ .

Equation (4) is the *curvature memory relation*. The field  $\Psi(x)$  is a weighted integral over the past light cone of  $x$ : each past curvature event contributes according to its causal influence, attenuated by the field mass. The non-locality in (4) is not a modification of the variational principle—the action (1) is fully local—but a consequence of causality imposed as a boundary condition. Causality of the retarded kernel was verified numerically (Sim 82: pre-source amplitude  $< 2 \times 10^{-14}$ ; Sim 83: memory oscillation period  $T \approx 2\pi/m_{\text{eff}}$ ).

### 3.5 Gravitational Field Equation

Varying (1) with respect to  $g^{\mu\nu}$  and using the standard identity for a non-minimally coupled scalar,  $\delta(\sqrt{-g}f(\Psi)R)/\delta g^{\mu\nu} = \sqrt{-g}[fG_{\mu\nu} + g_{\mu\nu}\square f - \nabla_\mu \nabla_\nu f]$ , with  $G_{\text{eff}}$  defined in (5), yields

$$G_{\text{eff}}(\Psi) \equiv \frac{G}{1 + 16\pi G\Lambda(\Psi)}, \quad (5)$$

$$\boxed{G_{\mu\nu} = 8\pi G_{\text{eff}}(\Psi) [T_{\mu\nu}^{(\Psi)} + T_{\mu\nu}^{(m)} + 2\nabla_\mu \nabla_\nu \Lambda(\Psi) - 2g_{\mu\nu} \square \Lambda(\Psi)]}, \quad (6)$$

with the recursive field stress-energy tensor

$$T_{\mu\nu}^{(\Psi)} = \nabla_\mu \Psi \nabla_\nu \Psi - g_{\mu\nu} \left[ \frac{1}{2} (\nabla \Psi)^2 - \frac{1}{2} m(\Psi)^2 \Psi^2 - V(\Psi) \right]. \quad (7)$$

Equation (6) replaces the phenomenological  $\mathcal{G}_{\mu\nu} = \mathcal{F}_{\mu\nu} + \mathcal{M}_{\mu\nu}$  of the v1 draft. The memory structure is now derived, not postulated:  $T_{\mu\nu}^{(\Psi)}$  sources geometry with an effective mass-energy distribution that is non-local in time because  $\Psi$  carries curvature history through (4); the terms  $2\nabla_\mu \nabla_\nu \Lambda - 2g_{\mu\nu} \square \Lambda$  encode gradients of the memory coupling and vanish when  $\Lambda = \text{const}$ .

### 3.6 FLRW Reduction and Modified Friedmann Equations

In spatially flat FLRW with homogeneous  $\Psi(t)$ ,  $R = 6(\dot{H} + 2H^2)$  and  $\square \Psi = \ddot{\Psi} + 3H\dot{\Psi}$ . The scalar equation (3) becomes

$$\ddot{\Psi} + 3H\dot{\Psi} + m_{\text{eff}}^2(\Psi) \Psi = 6\Lambda'(\Psi) (\dot{H} + 2H^2) + V'(\Psi). \quad (8)$$

The (00)-component of (6) yields the modified Friedmann equation

$$\boxed{3H^2 = 8\pi G_{\text{eff}}(\Psi) \left[ \rho_m + \frac{1}{2} \dot{\Psi}^2 + \frac{1}{2} m(\Psi)^2 \Psi^2 + V(\Psi) - 6H \Lambda'(\Psi) \dot{\Psi} \right]}. \quad (9)$$

The term  $-6H\Lambda'\dot{\Psi}$  is the *memory feedback contribution*. Its sign is fixed by the covariant derivation: for  $\Lambda' > 0$  and  $\dot{\Psi} > 0$ , memory suppresses the effective energy density driving expansion. The v1 coefficient was  $+3H\Lambda'\dot{\Psi}$  (sign wrong, factor-of-two wrong); the correct  $-6H\Lambda'\dot{\Psi}$  was derived analytically and confirmed numerically (Sim 85: Friedmann constraint residual  $\leq 3 \times 10^{-16}$ ; radiation-era scaling  $H \propto a^{-1.984}$ , cf. GR  $a^{-2}$ ).

## 4 Recursive Field Equations (Scalar Attractor)

The scalar field equation (3) in curved spacetime is the *attractor equation* of RIFT. It governs how memory and curvature interact nonlinearly to evolve the geometry. Physical interpretation:

- $\square \Psi$ : standard covariant wave propagation;
- $m_{\text{eff}}^2(\Psi) \Psi$ : nonlinear restoring force (recursive mass feedback);
- $\Lambda'(\Psi) R$ : curvature source term—the geometric driving of field growth;
- $V'(\Psi)$ : attractor gradient toward stable field configuration.

Together, these drive  $\Psi$  toward stable configurations that encode past curvature and suppress divergence. The field becomes both the memory *and* architect of spacetime.

## Boundary Conditions.

- *Low-memory limit* ( $m \rightarrow m_0$ ,  $\Lambda \rightarrow \text{const}$ ): recovers standard Klein-Gordon in curved spacetime.
- *High-curvature regime*: feedback becomes dominant, enabling emergent gravitational effects—flat rotation curves, void lensing, halo stabilization—without dark matter.
- *Late-time dynamics*:  $\Psi$  relaxes into attractor basins defined by  $V(\Psi)$ , forming stable geometric structures.

## 5 Canonical Quantization

To move from classical recursion to quantum curvature, we quantize the scalar field  $\Psi$ , its memory contribution, and the curvature ripple modes  $h_{\mu\nu}$ .

### 5.1 Scalar Field Quantization

The conjugate momentum is  $\Pi(x) = \partial_0 \Psi(x)$ . Equal-time commutator:  $[\Psi(\vec{x}, t), \Pi(\vec{y}, t)] = i\delta^3(\vec{x} - \vec{y})$ . Field expansion:

$$\Psi(x) = \int \frac{d^3k}{(2\pi)^3 \sqrt{2\omega_k}} \left( a_{\mathbf{k}} e^{ikx} + a_{\mathbf{k}}^\dagger e^{-ikx} \right), \quad (10)$$

with effective mass  $m_{\text{eff}}(\Psi)$  including feedback from curvature memory.

### 5.2 Tensor Field Quantization (Recursive Graviton)

Define the traceless symmetric perturbation  $h_{\mu\nu}$ . The wave equation is

$$\square h_{\mu\nu} + \alpha(\Psi) R h_{\mu\nu} = 0, \quad (11)$$

with equal-time commutator  $[h_{\mu\nu}(x), \partial_0 h_{\rho\sigma}(y)] = i\delta^3(x - y) \mathcal{P}_{\mu\nu, \rho\sigma}$ , where the polarization tensor  $\mathcal{P}$  enforces spin-2 structure.

### 5.3 Recursive Memory Decoherence and UV Finiteness

Memory tensor commutators decay as  $[\mathcal{M}_{\mu\nu}(x), \mathcal{M}_{\rho\sigma}(y)] \propto e^{-|t_x - t_y|/\tau_m}$ , suppressing UV contributions. This commutator decay *suggests* that vacuum fluctuations are damped at high frequency and loop diagrams may converge without explicit renormalization. However, no one-loop calculation has been performed to verify this; UV finiteness of the interacting theory beyond the free-field level remains a conjecture (noted in the abstract). This structural motivation distinguishes RIFT from standard QFT on a fixed background but does not yet constitute a proof.



**Note on nonlinear coupling.** The canonical quantization in Sections 5 applies to the free scalar and tensor sectors. The nonlinear interaction  $\Lambda(\Psi) R \Psi$  introduces cubic and higher interaction vertices in the quantum theory. Loop corrections to the propagator and vertex functions arising from this Ricci-coupling have not been computed. In particular, whether the interacting theory is UV-finite at loop level, and whether the memory-sector commutator decay extends to the interaction vertices, are open questions requiring dedicated perturbative calculation.

## 6 CMB Power Spectrum from Recursive Fields

### 6.1 RIFT Primordial Power Spectrum

The recursive field generates a primordial power spectrum that enters the CLASS Boltzmann solver as an external  $P(k)$ :

$$\mathcal{P}_\Psi(k) = A_s \left( \frac{k}{k_*} \right)^{n_s-1} \cdot \mathcal{A}_{\text{rec}}(k) \cdot \mathcal{D}_{\text{mem}}(k), \quad (12)$$

where  $\mathcal{A}_{\text{rec}}(k) = 1 + \epsilon \log(1 + k/k_c)$  is a recursive amplification factor from curvature feedback, and  $\mathcal{D}_{\text{mem}}(k) = \exp(-k^2/k_m^2)$  suppresses small-scale modes via memory decay.

In the full CLASS computation (Sim 88), the RIFT growth factor modification from  $G_{\text{eff}}$  is negligible at  $\Lambda_0 = 0.003$  ( $G_{\text{eff}}/G = 1 - 16$  ppm,  $D_{\text{RIFT}}(z=0)/D_{\Lambda\text{CDM}}(z=0) = 1.000000$  to six decimal places). The dominant RIFT CMB signal is therefore the acoustic peak shift driven by the modified background cosmology (Section 7.2).

### 6.2 BAO from Recursive Echo Shells

In  $\Lambda\text{CDM}$ , BAO arise from sound waves in the early photon-baryon plasma. In RIFT, analogous large-scale structure is *interpreted* as arising from *recursive echo shells*: as  $\Psi$  evolves, it emits ripple fronts when memory or curvature gradients reach critical levels. These waves stall at fixed radii due to recursive damping, form shell-like overdensities, and mimic BAO without requiring baryon pressure. Shell locations are set by  $r_n = v_c \cdot \tau_n$  where  $v_c$  is the recursive coherence velocity and  $\tau_n$  is the time to recursive saturation.

*Important caveat:* this is a physical interpretation, not a first-principles derivation of the sound horizon. The BAO scale  $r_d \approx 147$  Mpc is a free parameter fitted to data in Sim 87; no RIFT-specific prediction of  $r_d$  has been derived from the field equations. At  $\Lambda_0 \ll 1$  the RIFT background is indistinguishable from the  $\Lambda\text{CDM}$  FLRW background, so both models share the same sound horizon by construction. The echo-shell picture provides a qualitative description of *why* a BAO-like feature might arise in RIFT, but it does not constitute a derivation of the shell spacing.

Observationally, the BAO scale  $r_d \approx 147$  Mpc is reproduced by the RIFT best-fit cosmology, tested against BOSS DR12, eBOSS DR16, and Ly- $\alpha$  data with full covariance (Section 7.1).

## 7 Observational Consistency Tests

This section reports the four precision cosmological tests of RIFT (Simulations 87–90). All simulations are deterministic, reproducible, and publicly archived at `Ordered.Simulations/SIM87--SIM`

## 7.1 BAO Full-Covariance Refit (Sim 87)

**Problem with v1.** The prior BAO chi-squared (sim13\_4) used diagonal errors only:  $\chi_{\text{diag}}^2 = \sum_i (d_i - m_i)^2 / \sigma_i^2$ . BOSS DR12 and eBOSS DR16 publish off-diagonal DM–DH correlation coefficients  $\rho \approx -0.5$ . These are physically meaningful: DM and DH are both derived from the same BAO peak position, so a positive shift in DM is partially compensated by a negative shift in DH. Ignoring the correlation inflates  $\chi^2$ .

**Method.** We assembled the full  $12 \times 12$  covariance matrix from published off-diagonal correlation coefficients (Table 1), inverted via Cholesky decomposition, and computed  $\chi_{\text{full}}^2 = (\mathbf{d} - \mathbf{m})^\top C^{-1} (\mathbf{d} - \mathbf{m})$ . The data vector comprises 12 measurements: DM/ $r_d$  and DH/ $r_d$  at  $z = 0.38, 0.51, 0.70, 1.48, 2.33$ , plus DV/ $r_d$  at  $z = 0.15$  (6dFGS+MGS) and  $z = 0.85$  (eBOSS ELG). The RIFT background model integrates the full FLRW+scalar system (equations (8)–(9)) numerically via RK45. We optimised over  $(H_0, \Omega_m, r_d, \Lambda_0)$  with Nelder-Mead (20 random restarts).

Table 1: Published DM–DH correlation coefficients used in the BAO covariance matrix.

| Survey            | Redshift   | $\rho(\text{DM, DH})$ | Reference                       |
|-------------------|------------|-----------------------|---------------------------------|
| BOSS DR12         | $z = 0.38$ | $-0.52$               | Alam et al. (2017), Table 4     |
| BOSS DR12         | $z = 0.51$ | $-0.47$               | Alam et al. (2017), Table 4     |
| eBOSS DR16 LRG    | $z = 0.70$ | $-0.48$               | Alam et al. (2021), Table 3     |
| eBOSS DR16 QSO    | $z = 1.48$ | $-0.46$               | Alam et al. (2021), Table 3     |
| Ly- $\alpha$ DR16 | $z = 2.33$ | $-0.43$               | Bourboux et al. (2020), Table 3 |

## Results.

$$\begin{aligned}
\chi_{\text{diag, best}}^2 &= 11.74 \quad (\chi^2/\text{dof} = 1.47) \quad [\text{incorrect}] \\
\chi_{\text{full, best}}^2 &= \mathbf{9.78} \quad (\chi^2/\text{dof} = \mathbf{1.22}) \quad [\text{correct}] \\
\Delta\chi^2 &= +1.95 \quad (\text{full-cov preferred})
\end{aligned}$$

Best-fit parameters:  $\Omega_m = 0.294$ ,  $\Lambda_0 = 0.003$ ,  $H_0 \cdot r_d \approx \text{const}$  ( $H_0$  and  $r_d$  individually degenerate in BAO-only fits; only  $H_0 r_d$  is constrained). The  $\chi^2/\text{dof} = 1.22$  result should be reported to referees; the diagonal result is statistically incorrect.

## 7.2 CMB Full CLASS Injection (Sim 88)

**Problem with v1.** The prior CMB result used a first-order modulation  $C_\ell^{\text{RIFT}} \approx C_\ell^{\Lambda\text{CDM}}(1 + \epsilon)$ , bypassing the full CLASS Boltzmann hierarchy and missing the acoustic-peak shift from the modified background cosmology.

**Method.** We derived the RIFT primordial power spectrum  $P(k) = A_s(k/k_*)^{n_s-1} \cdot R_{\text{growth}}^2$  from the RIFT growth factor  $D_{\text{RIFT}}(z)/D_{\Lambda\text{CDM}}(z)$ , integrated via the modified linear growth equation with  $G_{\text{eff}}(\Psi)$ . This  $P(k)$  was injected into CLASS as `external_Pk` (format:  $k$  [Mpc $^{-1}$ ],  $\Delta_R^2(k)$  dimensionless) and the full Boltzmann transfer function was computed for both RIFT ( $H_0 = 68.14$ ,  $\Omega_m = 0.294$ ,  $\Lambda_0 = 0.003$ ) and  $\Lambda\text{CDM}$  (Planck 2018 best-fit:  $H_0 = 67.36$ ,  $\Omega_m = 0.3153$ ).

## Results.

$$\begin{aligned} G_{\text{eff}}/G &= 0.999984 \quad (16 \text{ ppm}) \\ D_{\text{RIFT}}(z=0)/D_{\Lambda\text{CDM}}(z=0) &= 1.000000 \quad (\text{to } 6 \text{ d.p.}) \\ \text{RMS}(\Delta C_\ell/C_\ell) &= \mathbf{2.93\%} \quad (l = 2\text{--}1500) \end{aligned}$$

The dominant RIFT CMB signal is the acoustic peak shift from the modified background ( $H_0 = 68.14$  vs.  $67.36$ ,  $\Omega_m = 0.294$  vs.  $0.315$ ), not the  $G_{\text{eff}}$  correction (16 ppm). The 2.93% RMS deviation is within Planck error bars at most multipoles.

### 7.3 Planck TT Likelihood (Sim 89)

**Method.** We implemented the approximate full-sky Gaussian TT likelihood

$$-2 \ln \mathcal{L} = f_{\text{sky}} \sum_{\ell} (2\ell + 1) [x_{\ell} - 1 - \ln x_{\ell}], \quad x_{\ell} = \frac{D_{\ell}^{\text{pred}} + N_{\ell}}{D_{\ell}^{\text{Planck}} + N_{\ell}}, \quad (13)$$

against the Planck 2018 best-fit TT theory spectrum (COM.PowerSpect.CMB-base-plikHM-TTTEEE R3.01), with  $f_{\text{sky}} = 0.778$  (Planck plikHM TT) and Planck 143 GHz noise  $N_{\ell}$  ( $\Delta_T = 33 \mu\text{K} \cdot \text{arcmin}$ ,  $\theta_{\text{FWHM}} = 7.27'$ ). Note:  $x_{\ell} = 1$  when  $D_{\ell}^{\text{pred}} = D_{\ell}^{\text{Planck}}$ , so  $-2 \ln \mathcal{L} = 0$  at the reference cosmology; the  $\Lambda\text{CDM}$  baseline  $\chi^2 = 6.2$  validates the implementation.

## Results.

| Model                                         | $\chi^2$ | Notes                       |
|-----------------------------------------------|----------|-----------------------------|
| $\Lambda\text{CDM}$ (CLASS vs. Planck theory) | 6.2      | Baseline; validates formula |
| RIFT (CLASS vs. Planck theory)                | 913      | BAO-only best-fit params    |
| $\Delta\chi^2$ (RIFT- $\Lambda\text{CDM}$ )   | +907     | $\sim 30.1\sigma$           |

The  $30.1\sigma$  tension reflects that the RIFT BAO-only best-fit ( $H_0 = 68.14$ ,  $\Omega_m = 0.294$ ) predicts acoustic peaks shifted from the Planck  $\Lambda\text{CDM}$  spectrum. The per-mode  $\chi^2 = 1.9 \times 10^{-4}$  is tiny; the large total arises from coherent accumulation over  $\sim 4.9 \times 10^6$   $f_{\text{sky}}$ -weighted modes. This is a parameter-consistency finding, not a failure of the RIFT field equations. Its resolution is the joint fit (Section 7.4).

### 7.4 Joint CMB+BAO Parameter Fit (Sim 90)

**Method.** We minimised  $\chi_{\text{total}}^2 = \chi_{\text{CMB}}^2(H_0, \Omega_m) + \chi_{\text{BAO}}^2(H_0, \Omega_m, r_d, \Lambda_0)$  jointly over all parameters. The CMB chi-squared uses the Gaussian approximation from the Planck 2018 published parameter covariance (Table 2 of [Planck Collaboration 2020](#)):  $\chi_{\text{CMB}}^2 = (\mathbf{x} - \boldsymbol{\mu})^\top C_{\text{Planck}}^{-1} (\mathbf{x} - \boldsymbol{\mu})$  with  $\boldsymbol{\mu} = (67.36, 0.3153)$ ,  $\sigma_{H_0} = 0.54$ ,  $\sigma_{\Omega_m} = 0.0073$ ,  $\rho = -0.90$ . The BAO chi-squared uses the same 12-point data vector and  $12 \times 12$  covariance matrix as Sim 87, evaluated with the flat Friedmann approximation for  $H(z)$  including the first-order  $G_{\text{eff}}$  correction at representative  $\Psi_0 = 0.01$  (identical method to Sim 91; the full ODE result agrees to  $< 16$  ppm per Sim 88). LCDM:  $\Lambda_0$  fixed to zero. RIFT:  $\Lambda_0$  free. Nelder-Mead optimisation, 20 random restarts.

## Results.

| Model               | $H_0$ | $\Omega_m$ | $r_d$  | $\Lambda_0$             | $\chi^2_{\text{CMB}}$ | $\chi^2_{\text{BAO}}$ |
|---------------------|-------|------------|--------|-------------------------|-----------------------|-----------------------|
| $\Lambda\text{CDM}$ | 67.59 | 0.3118     | 147.56 | 0                       | 0.23                  | 11.02                 |
| RIFT                | 67.59 | 0.3118     | 147.57 | $\approx 0.013^\dagger$ | 0.23                  | 11.02                 |

<sup>†</sup> Numerically unconstrained:  $\Delta\chi^2 < 0.004$  across  $\Lambda_0 \in [0, 0.1]$  (Sim 91). Value depends on optimiser init.

$$\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = \mathbf{0.000} \quad (\text{PASS})$$

The Planck CMB prior dominates the joint fit ( $\sigma_{H_0} = 0.54$  km/s/Mpc is  $\sim 10\times$  more constraining than BAO alone). Both RIFT and  $\Lambda\text{CDM}$  converge to the same optimal  $(H_0, \Omega_m) \approx (67.6, 0.312)$ . The coupling  $\Lambda_0 = 0.003$  modifies  $H(z)$  at the 16 ppm level (Sim 88), making RIFT and  $\Lambda\text{CDM}$  cosmologically indistinguishable at the CMB+BAO level.  $\Delta\chi^2 \approx 0$  confirms that RIFT is consistent with precision cosmological data at current measurement precision, and that the Sim 89 tension was an artefact of using the BAO-only (parameter-degenerate) best-fit as the CMB input. It is important to interpret this result correctly:  $\Delta\chi^2 \approx 0$  arises because at the joint best-fit coupling ( $\Lambda_0 \lesssim 0.013$ ,  $G_{\text{eff}}/G = 1 - 16$  ppm) RIFT is observationally degenerate with  $\Lambda\text{CDM}$ —both models converge to the same parameters and the RIFT correction is below current measurement precision. This should not be read as RIFT providing a physically distinct fit to CMB data; rather, the Sim 90 joint fit demonstrates that the theory is *consistent* with the data, not that it is *preferred* or that it explains a feature not accounted for by  $\Lambda\text{CDM}$ .

*Note on  $\Lambda_0$  reporting.* Two  $\Lambda_0$  values appear in the analysis: the BAO-only best-fit  $\Lambda_0 = 0.003$  (Sim 87, four free parameters, physically motivated) and the joint-fit value  $\Lambda_0 \approx 0.013$  (Sim 90, numerically unconstrained because the  $\chi^2$  landscape is flat in  $\Lambda_0$  at this precision,  $\Delta\chi^2_{\Lambda_0} < 0.004$  across  $[0, 0.1]$  from Sim 91; the specific value depends on optimiser initialisation and carries no physical significance). The fiducial value quoted throughout is  $\Lambda_0 = 0.003$ ; both are consistent with all current data.

**Summary.** Table 2 collects the key observational results. Best-fit parameter values for all simulations are tabulated in Appendix A.

## 7.5 Non-linear Structure Growth and $\sigma_8$ (Sim 92)

**Method.** We computed the linear and non-linear matter power spectra for the SIM90 joint best-fit cosmology ( $H_0 = 67.59$  km/s/Mpc,  $\Omega_m = 0.312$ ,  $\Omega_b = 0.049$ ,  $n_s = 0.9649$ ,  $A_s = 2.101 \times 10^{-9}$ ) across 15 values of  $\Lambda_0 \in [0, 0.1]$ . The linear  $P(k)$  was generated by CLASS at  $z = 0$  with the RIFT modification  $P_{\text{RIFT}}(k) = P_{\Lambda\text{CDM}}(k) \times (D_{\text{RIFT}}/D_{\Lambda\text{CDM}})^2$ , where the growth factor ratio is taken from Sim 88. Non-linear corrections were applied via the Takahashi et al. (2012) HALOFIT fitting formula [Takahashi et al., 2012]. The matter fluctuation amplitude is

$$\sigma_8^2 = \int_0^\infty \frac{k^2 dk}{2\pi^2} P_{\text{lin}}(k) \left[ \frac{3j_1(k \cdot 8 h^{-1} \text{Mpc})}{k \cdot 8 h^{-1} \text{Mpc}} \right]^2, \quad (14)$$

computed on  $k \in [10^{-4}, 10^2] h \text{Mpc}^{-1}$  with 2000 log-spaced points. The weak-lensing amplitude  $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$  is evaluated at the joint best-fit  $\Omega_m = 0.312$ .

**Results.** At the  $\Lambda$ CDM baseline ( $\Lambda_0 = 0$ ), the joint best-fit cosmology gives  $\sigma_8^{\text{lin}} = 0.824$ , consistent with the Planck 2018 constraint  $\sigma_8 = 0.811 \pm 0.006$  [Planck Collaboration, 2020] at the  $2.2\sigma$  level (expected given our simplified CMB Gaussian prior). The RIFT coupling suppresses the linear growth factor, reducing  $\sigma_8$  proportionally to  $D_{\text{RIFT}}^2/D_{\Lambda\text{CDM}}^2$ :

| $\Lambda_0$ | $D_{\text{RIFT}}/D_{\Lambda\text{CDM}}$ | $\sigma_8^{\text{lin}}$ | $\Delta\sigma_8$ | $S_8$ |
|-------------|-----------------------------------------|-------------------------|------------------|-------|
| 0.000       | 1.000000                                | 0.8239                  | 0.000%           | 0.840 |
| 0.003       | 0.999931                                | 0.8238                  | −0.007%          | 0.840 |
| 0.008       | 0.999815                                | 0.8237                  | −0.018%          | 0.840 |
| 0.050       | 0.998845                                | 0.8229                  | −0.116%          | 0.839 |
| 0.100       | 0.997691                                | 0.8220                  | −0.231%          | 0.838 |

The Planck/KiDS-1000  $\sigma_8$  tension amounts to  $\sim 5\text{--}6\%$  in  $\sigma_8$  (KiDS-1000:  $S_8 = 0.766 \pm 0.020$ , Asgari et al. 2021; DES Y3:  $S_8 = 0.776 \pm 0.017$ , Abbott et al. 2022). The RIFT suppression at the BAO best-fit coupling ( $\Lambda_0 = 0.003$ ) is  $\Delta\sigma_8 = -0.007\%$ —two orders of magnitude too small to alleviate this tension. Even at the detection threshold ( $\Lambda_0 = 0.05$ ),  $\Delta\sigma_8 = -0.116\%$ , still  $50\times$  below the observed discrepancy.

**Conclusion.** RIFT modifies  $\sigma_8$  only through the linear growth factor. At physically motivated coupling strengths ( $\Lambda_0 \leq 0.05$ ), the modification is  $|\Delta\sigma_8| < 0.12\%$ , and the Planck/KiDS  $\sigma_8$  tension is *not alleviated*. This is a falsifiable prediction: RIFT makes no claim to resolve the  $\sigma_8$  tension at current observational precision.

## 7.6 DESI Year 1 BAO Refit (Sim 94)

**Motivation.** Sim 87 established RIFT consistency with the BOSS+eBOSS BAO dataset ( $\chi^2/\text{dof} = 1.22$ ). The DESI Collaboration has since released Year 1 BAO measurements [Adame et al., 2024] spanning 13 data points across 7 tracer bins ( $0.3 \leq z_{\text{eff}} \leq 2.33$ ), providing an independent cross-check on the RIFT fit and the joint CMB+BAO parameter solution.

**Data.** We use the DESI Y1 BAO measurements from Table 1 of Adame et al. [2024]: BGS ( $z_{\text{eff}} = 0.295$ ,  $D_V/r_d$ ); LRG ( $z = 0.51, 0.71$ ); LRG+ELG ( $z = 0.93$ ); ELG ( $z = 1.32$ ); QSO ( $z = 1.49$ ); Lya QSO ( $z = 2.33$ ), with the published within-bin DM–DH correlation coefficients  $\rho \in [-0.49, -0.39]$ . The covariance is block-diagonal (one  $2 \times 2$  block per paired bin, zero between bins), following the published survey geometry.

### Results.

- BAO-only fit:**  $\Lambda$ CDM achieves  $\chi^2/\text{dof} = 13.6/10 = 1.36$  on the DESI data; RIFT gives an identical  $\chi^2$  ( $\Delta\chi^2 = 0$ ), consistent with the finding (Sim 91) that  $\Lambda_0$  does not enter  $H(z)$  at detectable levels for  $\Lambda_0 \leq 0.1$ . The slightly elevated  $\chi^2/\text{dof}$  relative to BOSS (1.22) is driven by known 2–3 $\sigma$  tensions in the DESI LRG bins ( $z = 0.51$   $D_H/r_d$ :  $-2.5\sigma$ ;  $z = 0.71$   $D_M/r_d$ :  $-2.0\sigma$ ), a feature of the DESI data independent of the gravity model.
- Joint DESI+CMB fit:** Adding the Planck 2018 Gaussian prior, the joint minimum gives  $H_0 = 67.66$  km/s/Mpc,  $\Omega_m = 0.311$ ,  $r_d = 148.5$  Mpc,  $\Lambda_0 = 0.063$

(unconstrained);  $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) \approx 0$ . These parameters are consistent with the SIM90 BOSS-based joint fit ( $H_0 = 67.59$ ,  $\Omega_m = 0.312$ ) at the  $\lesssim 0.2\sigma$  level.

3.  **$\Lambda_0$  sensitivity:** The DESI BAO  $\chi^2$  is flat across  $\Lambda_0 \in [0, 0.1]$  ( $\Delta\chi_{\text{max}}^2 < 10^{-4}$ ), confirming that the RIFT field modification is undetectable at current BAO precision regardless of the dataset used.

**Conclusion.** RIFT passes the DESI Y1 BAO test with  $\Delta\chi^2 = 0$  relative to  $\Lambda\text{CDM}$ . The joint CMB+DESI parameter solution is fully consistent with the BOSS-based SIM90 result, confirming the stability of RIFT parameters across independent BAO datasets.  $\Lambda_0$  remains a null parameter at current observational precision.

## 7.7 CMB Polarization EE/TE (Sim 95)

**Motivation.** Simulations 88–89 established RIFT consistency with the Planck 2018 temperature (TT) power spectrum. Sim 95 extends this test to E-mode polarization: the EE auto-power spectrum and the TE cross-power spectrum. Polarization is independently generated at recombination and provides a complementary probe of the acoustic-peak structure, reionization optical depth, and matter content.

**Method.** We run the CLASS Boltzmann code with `output = tCl,pCl,lCl` for three cosmologies: (A) the RIFT BAO-only best-fit (SIM87:  $H_0 = 68.14$ ,  $\Omega_m = 0.294$ ), (B) the RIFT joint CMB+BAO best-fit (SIM90:  $H_0 = 67.59$ ,  $\Omega_m = 0.312$ ), and (C) the LCDM Planck 2018 best-fit baseline. The RIFT  $P(k)$  is constructed from the growth-factor ratio  $R = D_{\text{RIFT}}/D_{\Lambda\text{CDM}}$  (identical to Sim 88). The lensed  $C_\ell^{EE}$  and  $C_\ell^{TE}$  are compared against the Planck 2018 published best-fit theory spectrum (`COM_PowerSpect_CMB-base-plikHM-TTTEEE R3.01`) over  $\ell = 2\text{--}1996$ .

### Results.

1. **Growth factor:**  $D_{\text{RIFT}}(z=0)/D_{\Lambda\text{CDM}}(z=0) = 1.000000$  at both coupling values ( $\Lambda_0 = 0.003$  and  $0.013$ ). The  $G_{\text{eff}}$  correction ( $\lesssim 65$  ppm) is negligible in all polarization spectra.
2. **Joint best-fit RMS deviations** (RIFT vs LCDM Planck,  $\ell = 2\text{--}1996$ ):  $\text{RMS}(\Delta C_\ell^{EE}/C_\ell^{EE}) = 0.19\%$ ;  $\text{RMS}(\Delta C_\ell^{TE}/\text{RMS}(C_\ell^{TE})) = 0.43\%$ ;  $\text{RMS}(\Delta C_\ell^{TT}/C_\ell^{TT}) = 0.20\%$ . All are well below Planck’s measurement uncertainties at most multipoles. The small deviations arise from the  $(H_0, \Omega_m)$  offset between the RIFT joint best-fit and the Planck LCDM reference point.
3. **Approximate Planck polarization likelihood:** Using a Gaussian approximation over 1995 polarization modes, the LCDM baseline gives  $\chi_{\text{EE}}^2 = 57.6$ ,  $\chi_{\text{TE}}^2 = 37.2$ ; the RIFT joint best-fit gives  $\chi_{\text{EE}}^2 = 89.0$ ,  $\chi_{\text{TE}}^2 = 60.0$  ( $\Delta\chi_{\text{EE}}^2 = +31.5$ ,  $\Delta\chi_{\text{TE}}^2 = +22.8$ ). These excesses arise from the same parameter offset that generates the small TT deviation after the joint fit.
4. **BAO-only tension in polarization:** The RIFT BAO-only best-fit produces  $\chi_{\text{EE}}^2 = 4353$  and  $\chi_{\text{TE}}^2 = 3110$  vs Planck, confirming that the TT tension identified in Sim 89 ( $\Delta\chi_{\text{TT}}^2 = 907$ ) propagates to all CMB spectra equally. This is consistent with the tension being driven by the  $(H_0, \Omega_m)$  parameter-space shift, not by the RIFT field equations.



**Conclusion.** RIFT at the joint CMB+BAO best-fit reproduces Planck EE and TE with sub-percent fractional accuracy ( $< 0.2\%$  for EE,  $< 0.5\%$  for TE). The polarization results confirm: (i)  $G_{\text{eff}}$  modifications are invisible in polarization; (ii) the BAO-CMB tension is a parameter-space effect present in all spectra; (iii) the joint-fit parameter solution restores consistency across TT, EE, and TE simultaneously.

## 7.8 RSD Growth Rate $f\sigma_8$ (Sim 96)

Redshift-space distortions (RSD) measure the linear growth rate  $f(z) = d \ln D / d \ln a$  through the combination  $f\sigma_8(z)$ , where  $D(z)$  is the linear growth factor normalized to unity at  $z = 0$  and  $\sigma_{8,0} = 0.824$  is taken from the SIM90 joint best-fit. In RIFT the gravitational coupling governing growth is modified to  $G_{\text{eff}} = G/(1 + 16\pi\Lambda(\Psi))$ , entering the growth equation as

$$D'' + \left(2 + \frac{d \ln H}{d \ln a}\right) D' = \frac{3}{2} \frac{\Omega_m}{a^3 E^2} G_{\text{eff}} D, \quad (15)$$

where primes denote  $d/d \ln a$  and  $E(z) = H(z)/H_0$ .

**Data.** We use the gold-sample RSD compilation spanning  $z = 0.067$ – $1.48$  (Table 3): 6dFGRS [Beutler et al., 2012], SDSS MGS [Howlett et al., 2015], BOSS DR12 ( $z = 0.38, 0.51, 0.61$ ) [Alam et al., 2017], VIPERS [Pezzotta et al., 2017], eBOSS DR16 LRG [Gil-Marín et al., 2020], eBOSS DR16 ELG [de Mattia et al., 2021], and eBOSS DR16 QSO [Neveux et al., 2020]. All model predictions use the SIM90 cosmological background with no additional free parameters.

**Results.** At the SIM90 joint best-fit ( $H_0 = 67.59$ ,  $\Omega_m = 0.312$ ,  $\Lambda_0 \approx 0.013$ ) RIFT yields  $\chi^2/\text{dof} = 7.73/9 = 0.86$ , compared to  $7.87/9 = 0.87$  for  $\Lambda\text{CDM}$  Planck 2018, giving

$$\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = -0.14 \approx 0. \quad (16)$$

All individual pulls are within  $2\sigma$ . The maximum fractional deviation of the RIFT  $f\sigma_8$  curve from the  $\Lambda\text{CDM}$  curve at the best-fit is  $0.62\%$  (across  $z = 0$ – $2$ ), which is sub-percent and well below current RSD measurement precision.

**$G_{\text{eff}}$  suppression.** At  $\Lambda_0 \approx 0.013$  the correction  $16\pi\Lambda(\Psi) = 16\pi \times 0.013 \times (0.01)^2 \approx 6.5 \times 10^{-5}$ , so  $G_{\text{eff}}$  deviates from  $G$  by  $\approx 65$  ppm. The observed  $\sim 0.6\%$  difference in  $f\sigma_8$  between RIFT joint and  $\Lambda\text{CDM}$  arises entirely from the  $(H_0, \Omega_m)$  parameter offset, not from the recursive field.

**$\Lambda_0$  scan.** A scan of  $\Lambda_0 \in [0, 0.2]$  (holding all other parameters at SIM90 joint values) shows the  $f\sigma_8$  deviation from  $\Lambda\text{CDM}$  remains below  $1\%$  for  $\Lambda_0 < 0.2$ . The detection threshold exceeds the Bayesian upper limit  $\Lambda_0 < 0.095$  (95% CL, Sim 93), so current RSD data cannot independently constrain  $\Lambda_0$ .

**$S_8$  tension context.** With  $\sigma_{8,0} = 0.824$  and  $\Omega_m = 0.312$ , the RIFT prediction  $S_8 = \sigma_8 \sqrt{\Omega_m/0.3} = 0.840$  lies  $3.7\sigma$  above KiDS-1000 ( $S_8 = 0.766 \pm 0.020$ , Asgari et al. 2021), compared to  $3.9\sigma$  for  $\Lambda\text{CDM}$  Planck. RIFT provides no relief for the  $S_8$  tension, consistent with the SIM92 result ( $\Delta\sigma_8 < 0.007\%$  at  $\Lambda_0 = 0.003$ ).

**Conclusion.** RIFT passes all current RSD /  $f\sigma_8$  constraints. The growth rate is indistinguishable from  $\Lambda$ CDM at the joint best-fit because  $G_{\text{eff}} \approx G$  to better than 10 ppm. Detection of the RIFT growth signature requires  $\Lambda_0 > 0.2$ , which is excluded by the BAO+CMB joint fit, making the growth rate a consistency test rather than a discriminator with current data. Future surveys (DESI Y5, Euclid) measuring  $f\sigma_8$  to  $\sim 1\%$  at  $z > 1$  will tighten this constraint to  $\Lambda_0 \lesssim 0.05$ .

## 7.9 Definitive Planck 2018 plikHM TTTEEE Likelihood (Sim 97)

**Motivation.** Simulations 89 and 95 compared RIFT Boltzmann spectra against the Planck published theory curve using an approximate Gaussian CMB likelihood. Sim 97 replaces this with the *official* Planck 2018 plikHM TTTEEE likelihood code, accessed via the `clipy` v0.15 Python interface to the `clik` library [Planck Collaboration, 2020]. This is the identical likelihood used in the Planck 2018 cosmological parameter analysis.

**Method.** Three cosmologies are evaluated against the full official Planck 2018 likelihood: (A) RIFT joint CMB+BAO best-fit (Sim 90:  $H_0 = 67.59$ ,  $\Omega_m = 0.312$ ,  $\Lambda_0 = 0.008$ , original value at time of Sim 97 run); (B) RIFT BAO-only best-fit (Sim 87:  $H_0 = 68.14$ ,  $\Omega_m = 0.294$ ,  $\Lambda_0 = 0.003$ ); (C)  $\Lambda$ CDM Planck 2018 best-fit [Planck Collaboration, 2020].

For each cosmology, CLASS v3 computes the lensed TT, EE, and TE spectra with the RIFT background (same pipeline as Sim 95). The  $C_\ell$  vectors are evaluated by:

- **plikHM TTTEEE:** high- $\ell$  ( $\ell = 30$ –2508) likelihood with 47 nuisance parameters (foreground amplitudes, calibration) fixed at Planck best-fit values from the `clik` self-check vector;
- **lowl TT (Commander):** low- $\ell$  ( $\ell = 2$ –29) TT;
- **lowl EE (SimAll):** low- $\ell$  ( $\ell = 2$ –29) EE polarization.

*Pipeline validation.*  $\Lambda$ CDM gives  $-2 \ln \mathcal{L}_{\text{plikHM}} = 2352.05$ , compared to the `clik` self-check value  $2 \times 1172.47 = 2344.94$  (agreement to  $< 0.3\%$ , attributable to our parameter values not being bit-for-bit identical to the self-check cosmology).

### Results.

| Cosmology         | plikHM TTTEEE | lowl TT | lowl EE | Total $-2 \ln \mathcal{L}$ |
|-------------------|---------------|---------|---------|----------------------------|
| RIFT joint (A)    | 2359.05       | 23.54   | 396.05  | 2778.63                    |
| RIFT BAO-only (B) | 3297.80       | 24.26   | 395.93  | 3717.99                    |
| $\Lambda$ CDM (C) | 2352.05       | 23.42   | 396.14  | 2771.61                    |

$$\Delta(-2 \ln \mathcal{L})_{\text{plikHM}}(\text{RIFT joint} - \Lambda\text{CDM}) = +7.0 \quad (17)$$

$$\Delta(-2 \ln \mathcal{L})_{\text{plikHM}}(\text{RIFT BAO-only} - \Lambda\text{CDM}) = +945.7 \quad (18)$$



**Interpretation.** The BAO-only tension ( $\Delta(-2 \ln \mathcal{L}) = +946$ , Eq. 18) is fully consistent with the Sim 89 approximate result ( $\Delta\chi^2 = +907$ ), cross-validating both pipelines.

The RIFT joint best-fit (A) shows a  $\Delta(-2 \ln \mathcal{L}) = +7.0$  excess (Eq. 17). This arises from the  $(H_0, \Omega_m)$  offset between the Sim 90 joint parameters (optimised against the approximate Gaussian CMB prior) and the true Planck CMB optimum. At  $\Lambda_0 = 0.008$ ,  $G_{\text{eff}}/G = 1 - 42 \text{ ppm}$  contributes negligibly to any CMB observable; the  $+7.0$  excess is entirely parameter-driven. Specifically, RIFT evaluated at the Planck  $\Lambda\text{CDM}$  parameter point ( $H_0 = 67.36$ ,  $\Omega_m = 0.3153$ ,  $\Lambda_0 = 0.008$ ) yields  $\Delta(-2 \ln \mathcal{L}) \approx 0$ , since  $G_{\text{eff}}$  deviates from  $G$  by only 42 ppm.

In frequentist terms,  $\Delta(-2 \ln \mathcal{L}) = +7.0$  with one effective degree of freedom ( $\Lambda_0$  being the extra parameter) corresponds to  $\approx 2.6\sigma$  tension—a marginal result. This tension is resolved by Sim 98 below, which re-optimises the RIFT parameter fit using the real plikHM likelihood.

**Conclusion.** The RIFT *theory* is fully consistent with the real Planck 2018 plikHM TTTEEE likelihood: the  $G_{\text{eff}}$  correction at the joint best-fit coupling is 42 ppm and invisible to the CMB. The marginal  $\Delta(-2 \ln \mathcal{L}) = +7.0$  is a parameter-optimisation artefact that vanishes in Sim 98.

## 7.10 Joint plikHM + BAO Parameter Re-fit (Sim 98)

**Motivation.** Sim 97 revealed a  $\Delta(-2 \ln \mathcal{L}) = +7.0$  excess at the Sim 90 joint best-fit parameters. The cause was identified as the  $(H_0, \Omega_m)$  offset between Sim 90 (optimised against an approximate Gaussian CMB prior) and the true Planck CMB optimum. Sim 98 resolves this by re-running the joint CMB+BAO optimisation using the real plikHM TTTEEE likelihood.

**Method.** We minimise

$$\chi_{\text{total}}^2(H_0, \Omega_m, \Lambda_0) = -2 \ln \mathcal{L}_{\text{plikHM}} + (-2 \ln \mathcal{L}_{\text{lowlTT}}) + (-2 \ln \mathcal{L}_{\text{lowlEE}}) + \chi_{\text{BAO}}^2 \quad (19)$$

over  $(H_0, \Omega_m)$  for  $\Lambda\text{CDM}$  and  $(H_0, \Omega_m, \Lambda_0)$  for RIFT, using Nelder-Mead optimisation (max 80 CLASS evaluations each). BAO data: BOSS DR12 + eBOSS DR16 + Ly $\alpha$  with full  $12 \times 12$  covariance (same as Sim 87). The sound horizon  $r_d$  is computed analytically from  $(H_0, \Omega_m, \Omega_b)$  via numerical integration of  $c_s(a)/[a^2 H(a)]$  to the baryon drag epoch (Eisenstein & Hu 1998 approximation for  $z_{\text{drag}}$ ).

**Results.**

| Model               | $H_0$ | $\Omega_m$ | $\Lambda_0$ | $r_d$ | plikHM $-2 \ln \mathcal{L}$ | $G_{\text{eff}}/G$   |
|---------------------|-------|------------|-------------|-------|-----------------------------|----------------------|
| $\Lambda\text{CDM}$ | 67.30 | 0.3169     | 0           | 150.6 | 2347.46                     | 1                    |
| RIFT                | 67.30 | 0.3168     | 0.00506     | 150.6 | 2347.45                     | $1 - 24 \text{ ppm}$ |

$$\Delta(-2 \ln \mathcal{L})_{\text{plikHM}}(\text{RIFT} - \Lambda\text{CDM}) = -0.013 \approx 0 \quad (\text{PASS}) \quad (20)$$

The RIFT and  $\Lambda\text{CDM}$  best-fits are essentially identical:  $\Delta H_0 = +0.002$ ,  $\Delta \Omega_m = -0.0001$ ,  $\Delta \Lambda_0 = 0.005$  (with  $G_{\text{eff}}$  deviation 24 ppm). The  $\chi_{\text{total}}^2$  difference is  $+0.004$ .

## Progression of CMB+BAO tests.

| Simulation               | $\Delta(-2 \ln \mathcal{L})_{\text{plikHM}}$ | Parameter point       | Notes                 |
|--------------------------|----------------------------------------------|-----------------------|-----------------------|
| Sim 89 (approx Gaussian) | +907                                         | BAO-only best-fit     | Parameter tension     |
| Sim 90 (approx Gaussian) | $\approx 0$                                  | Joint approx best-fit | Resolved by joint fit |
| Sim 97 (real plikHM)     | +7.0                                         | Sim 90 approx params  | Residual offset       |
| Sim 98 (real plikHM)     | <b>-0.013</b>                                | Real plikHM best-fit  | <b>Fully resolved</b> |

**Conclusion.** When RIFT parameters are optimised against the real Planck 2018 plikHM TTTEEE likelihood,  $\Delta(-2 \ln \mathcal{L}) = -0.013 \approx 0$ . The Sim 97 tension of +7.0 is completely resolved. RIFT is fully consistent with the official Planck 2018 CMB likelihood at the optimised parameter point, with  $G_{\text{eff}}/G = 1 - 24 \text{ ppm}$ .

## 8 Recursive Graviton Emission

In RIFT, gravitational waves are emitted not from accelerated masses, but from curvature memory spikes that surpass feedback thresholds. These events generate quantized ripple packets: *recursive gravitons*.

Recursive graviton emission occurs when the memory tensor spike exceeds the threshold

$$\delta \mathcal{M}_{\mu\nu} \geq \frac{1}{\eta_c} \frac{d}{d\tau} (\nabla_{(\mu} \phi_{\nu)}), \quad (21)$$

where  $\phi_\mu$  is the recursive vector potential from  $\Psi$ ,  $\eta_c$  is the memory coupling constant, and  $\tau$  is proper time.

Each recursive graviton carries energy  $E = \hbar\omega \int |\delta \mathcal{M}_{\mu\nu}|^2 d^3x$ .

**Physical features.** Emission is discrete (triggered when curvature memory spikes), spin-2 symmetry is preserved, propagation slows in memory-rich zones due to decoherence, and the source is geometric rather than energetic.

**Testable predictions.** Echo-phase ripples in void regions, B-mode polarization with recursive ring harmonics, lensing interference from memory collapse (not binary mergers), nonthermal gravitational bursts unaligned with mass-based models.

## 9 Falsifiability Predictions

RIFT stands apart from speculative models by offering clear, testable predictions with well-defined thresholds for falsification.

### 9.1 Galaxy Rotation Curves

*Prediction:* Flat curves arise from recursive curvature—not dark matter. *Sim 99–100 status:* The locked action at  $\Lambda_0 = 0.003$  *does not* produce flat rotation curves via  $T^\Psi$  or  $G_{\text{eff}}$  in either the static (Sim 99) or time-domain (Sim 100) treatment. The decaying Yukawa profile ( $m_0 = 0.1 \text{ kpc}^{-1}$ ) is exponentially steeper inward than the  $1/r^2$

isothermal profile required for curve flattening (Section ??). The galactic Ricci coupling  $2\Lambda_0|R_{\text{gal}}| \approx 5 \times 10^{-9} \text{ kpc}^{-2}$  is  $5 \times 10^7$  times weaker than  $m_0^2$ : curvature-driven condensation is ruled out at galactic densities. Attractive self-coupling ( $\lambda < 0$ ) either leaves the Yukawa profile unchanged (sub-threshold) or causes tachyonic blow-up (super-threshold), with no stable halo condensate in between (Section ??). The DM-replacement prediction therefore requires derivation of a galactic-scale screening or higher-order stabilization mechanism not yet present in the locked action. *Test*: Reconstruct  $\Psi(r)$  from observed curves once such a mechanism is derived; verify that field gradient sustains rotation. *Falsification*: If no such mechanism can be derived from the locked action, the DM-replacement prediction is falsified.

## 9.2 Void Lensing (DES, Euclid)

*Prediction*: Weak lensing without mass near voids, via phase-delayed geodesics from residual memory  $\mathcal{M}_{\mu\nu}$ . *Test*: Cross-correlate Planck lensing with DES voids; look for phase shifts. *Falsification*: No lensing where RIFT predicts curvature  $\Rightarrow$  falsification.

## 9.3 Bullet Cluster Lensing Offset

*Prediction*: Lensing offset explained by curvature memory persisting after mass displacement. *Test*: Simulate lensing profiles with recursive memory; compare to  $\kappa$  maps. *Falsification*: Cannot reproduce lensing peak without dark mass.

## 9.4 CMB Low- $\ell$ Anomalies

*Prediction*: Recursive memory decay suppresses low- $\ell$  power. *Test*: Compare RIFT  $C_\ell$  to Planck 2018. Validate suppression shape. *Falsification*: Cannot match low- $\ell$  drop without fine-tuning.

# 10 Solar System Viability: Effective Matter Coupling (Sim 107)

## 10.1 Motivation

Passing CMB and BAO tests is necessary but not sufficient for a modified-gravity theory. The  $\Psi$  field mediates a long-range scalar force that must not violate Solar System constraints. We derive the  $\Psi$ -matter coupling from the linearised RIFT field equation and test against the Cassini bound ( $|\gamma - 1| < 2.3 \times 10^{-5}$ , Bertotti et al. 2003).

## 10.2 Effective Coupling

Linearising around background  $\Psi_0$  in the presence of matter perturbation  $\delta\rho$ :

$$\square \delta\Psi - m_{\text{eff}}^2(\rho) \delta\Psi = 16\pi G_N \Lambda_0 \Psi_0 \delta\rho, \quad m_{\text{eff}}^2(\rho) = m_0^2 + 2\Lambda_0 \rho.$$

In scalar-tensor language the dimensionless coupling is

$$\alpha(\Psi_0) = \frac{\Lambda_0 \Psi_0}{1 + 2\Lambda_0 \Psi_0^2}, \quad \gamma - 1 = \frac{-2\alpha^2}{1 + \alpha^2}.$$

The effective mass is *chameleon-like*:  $m_{\text{eff}}$  increases with density, screening the field inside compact bodies. At Solar interior density ( $\rho_{\odot} \approx 1400 \text{ kg m}^{-3}$ ) the Compton wavelength drops to  $\lambda_{\Psi} \approx 8 \times 10^{-11} \text{ Mpc}$  (sub-femtometre), providing complete screening. In interplanetary space  $\lambda_{\Psi} \sim 4 \times 10^6 \text{ Mpc}$ , so the force is unscreened but couples only as  $\alpha^2 \sim \Lambda_0^2 \Psi_0^2$ .

## 10.3 Results

At the natural background value  $\Psi_0 = 1 M_{\text{Pl}}$ :

$$\alpha^2 = 8.9 \times 10^{-6},$$

$$|\gamma - 1| = 1.78 \times 10^{-5} < 2.3 \times 10^{-5} \text{ (Cassini)}.$$

The critical constraint is  $|\Psi_0| < 1.14 M_{\text{Pl}}$ ; the locked action *requires no additional screening mechanism*. Table 5 summarises the environmental chameleon scan.

**Verdict.** RIFT passes Solar System viability for all  $|\Psi_0| \lesssim 1 M_{\text{Pl}}$ . The coupling is naturally suppressed by  $\alpha^2 = O(\Lambda_0^2) \sim 10^{-5}$ ; no fine-tuning is required.

# 11 Gravitational Wave Speed, GW170817, and BBN (Sim 108)

## 11.1 GW speed from the tensor perturbation equation

The transverse-traceless tensor perturbation of the RIFT action obeys

$$G_4(t) \left[ \ddot{h} + \left( 2H + \frac{\dot{G}_4}{G_4} \right) \dot{h} + \frac{k^2}{a^2} h \right] = 0, \quad G_4 = \frac{1}{16\pi G_N} + \Lambda_0 \Psi_0^2. \quad (22)$$

The gravitational wave speed  $c_T^2 = 1$  *exactly*: RIFT belongs to the Horndeski class with  $G_5 = 0$ , and  $G_5 \neq 0$  is the only source of  $c_T \neq c$  in that class.

## 11.2 Results

The  $G_{\text{eff}}$  constant offset of  $-0.60\%$  arises from  $G_{\text{eff}} = G_N / (1 + 2\Lambda_0 \Psi_0^2)$  at the observationally constrained  $\Lambda_0 = 0.003$ ,  $\Psi_0 \sim 1$ ; this is stable across all redshifts because the slow-roll approximation gives  $\Psi_0(z) \approx \Psi_0(0)$  for  $m_0 = 0.001 H_0$  throughout the observable universe.

# 12 Conclusion

Recursive Intelligence-Field Theory (RIFT) presents a new paradigm in gravitational physics: curvature is not statically sourced by mass, but dynamically shaped by memory.

### Key results.

- Unified Lagrangian (eq. (1)) couples mass, curvature, and entropy through a single field  $\Psi$ .

- All field equations—scalar attractor, gravitational equation, Friedmann system—are *derived*, not postulated.
- Numerical verification (Sims 82–86): causality (pre-source amplitude  $< 2 \times 10^{-14}$ ), stability ( $\omega^2 > 0$ ), GR recovery ( $\Psi = 0$  stable fixed point), Friedmann constraint  $\leq 3 \times 10^{-16}$ .
- Observational consistency (Sims 87–98): BAO  $\chi^2/\text{dof} = 1.22$  (BOSS, Sim 87), 1.36 (DESI Y1, Sim 94); CMB TT RMS = 2.93% (Sim 88); joint CMB+BAO  $\Delta\chi^2 = 0$  (both datasets);  $\sigma_8$  unmodified  $|\Delta\sigma_8| < 0.007\%$ ; EE/TE RMS  $< 0.2\%/0.5\%$  at joint best-fit (Sim 95); RSD  $f\sigma_8$ :  $\chi^2/\text{dof} = 0.86$ ,  $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = -0.14$  (Sim 96); official Planck plikHM:  $\Delta(-2\ln\mathcal{L}) = -0.013$  at optimum (Sim 98).
- Galaxy rotation curve tests (Sims 99–100, 103): the locked action at  $\Lambda_0 = 0.003$  does not replace dark matter. Static test (Sim 99): Yukawa halo exponentially steeper than the  $1/r^2$  isothermal profile required for flat curves. Time-domain test (Sim 100): the galactic curvature coupling  $2\Lambda_0|R_{\text{gal}}| \approx 5 \times 10^{-9} \text{ kpc}^{-2}$  is  $5 \times 10^7$  times weaker than  $m_0^2$ —curvature condensation is impossible at galactic densities. Vainshtein screening (Sim 103): the galactic field gradient  $|\delta\Psi'| \sim 10^{-8} \text{ kpc}^{-1}$  is  $10^6 \times$  below the nonlinear threshold for all scanned  $(\alpha_V, M_V)$ ; energy density deficit is  $2.7 \times 10^6 \times$  the isothermal requirement. All kinetic modifications of the locked action are exhausted; new physics (a first-principles coupling between  $\Psi$  and baryonic matter) is required.
- RG flow (Sim 105):  $\Lambda_0$  is asymptotically free — beta function is negative,  $\Lambda_0 \rightarrow 0$  as  $k_m \rightarrow \infty$ , UV fixed point is pure GR. Analytic running:  $\Lambda_0(k_m)^{-2} = \Lambda_0^{-2} + (k_m^2 - k_{m,\text{nat}}^2)/(16\pi^2 m_0^2)$ , confirmed numerically to  $< 0.001\%$ . IR running  $< 2.5\%$ ;  $\Lambda_0 = 0.003$  is an IR attractor; no Landau pole.
- Two-loop graviton sector (Sim 106): Ward identity  $\Pi_{hh}(0) = 0$  exact at all loop orders; resummed  $\Psi$  propagator regulates mixed diagrams;  $\delta Z_h \sim 10^{-5}$  negligible. Combined with Sims 102, 104, 105: RIFT UV-finite through two-loop graviton sector under Dyson resummation.
- UV regulation (Sims 102, 104): one-loop  $\Psi$  self-energy from  $\Lambda_0\Psi R$  vertex confirmed finite under memory damping —  $\Sigma(0) = \Lambda_0^2 k_m^4/(64\pi^2)$ ; quartic UV divergence regulated; naturalness requires  $\tau_m > 0.11 \text{ Mpc}$  (Sim 102). Full one-loop calculation (Sim 104): graviton wavefunction renormalization protected by diffeomorphism Ward identity  $\Pi_{hh}(0) = 0$ ; vertex correction  $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$  finite; two-loop  $\Sigma_D/m_0^2 = 3.4 \times 10^{-6}$  (perturbative hierarchy  $\Sigma_A \gg \Sigma_C \gg \Sigma_D$  confirmed). UV-finiteness holds at full one-loop level in the resummed theory; two-loop graviton self-energy remains open.
- Predicts recursive graviton emission via curvature memory spikes.
- BAO sound horizon from first principles (Sim 101):  $|\Delta r_d/r_d| < 21 \text{ ppm}$  across  $\Lambda_0 \in [0, 0.095]$ ; RIFT and LCDM degenerate in  $r_d$  at all allowed couplings. Echo-shell mechanism confirmed as physical interpretation of standard sound horizon physics.

**Honest assessment.** At current observational precision, RIFT at its cosmological best-fit ( $\Lambda_0 \approx 0.003\text{--}0.013$ ) is *observationally indistinguishable from  $\Lambda$ CDM* in every dataset tested: BAO (BOSS + DESI Y1), CMB TT/EE/TE, RSD  $f\sigma_8$ , and the official Planck plikHM likelihood. The coupling  $\Lambda_0$  is not detected—it is consistent with zero across the entire range  $[0, 0.1]$  tested, with a 95% upper limit  $\Lambda_0 < 0.095$  that is prior-dominated (Sim 93). RIFT is currently a *consistent extension* of  $\Lambda$ CDM, not an empirically preferred alternative. Its scientific value lies in its theoretical structure (a single derived Lagrangian, locked action, falsifiable predictions at  $\Lambda_0 \gtrsim 0.05$ ) and in explicitly ruling out naive claims (rotation curves do not arise from the locked action;  $\sigma_8$  tension is not alleviated), not in a superior fit to existing data.

**Coupling-strength constraints (Sim 91).** A dedicated sensitivity scan (Sim 91) mapped the observational signature of RIFT across  $\Lambda_0 \in [0, 0.1]$  at the joint best-fit cosmology ( $H_0 = 67.59$  km/s/Mpc,  $\Omega_m = 0.312$ ). Key results:

- $G_{\text{eff}}/G$  deviates from unity by at most 500 ppm at  $\Lambda_0 = 0.1$ ; current surveys require  $> 0.1\%$  sensitivity, placing the detection threshold beyond  $\Lambda_0 = 0.1$ .
- The linear growth factor ratio  $D_{\text{RIFT}}/D_{\Lambda\text{CDM}}$  exceeds 0.1% (structure-growth survey precision) at  $\Lambda_0 \gtrsim 0.05$ .
- The BAO chi-squared is flat across the full scan ( $\Delta\chi_{\text{BAO}}^2 = 0.004$  from  $\Lambda_0 = 0$  to 0.1)—RIFT  $H(z)$  is undetectable in current BAO data for any coupling in this range.
- $H(z)$  deviations remain below 0.03% RMS at  $\Lambda_0 = 0.1$ ; DESI-level precision ( $\sim 0.1\%$ ) would constrain  $\Lambda_0 \lesssim 0.1$ .

This confirms that RIFT at the BAO best-fit coupling ( $\Lambda_0 = 0.003$ ) is *cosmologically safe*—consistent with all current precision data—and establishes that current surveys constrain  $\Lambda_0 \lesssim 0.05$  from structure growth. Figure 7 summarises these results.

**Model selection (AIC/BIC).** Because RIFT and  $\Lambda$ CDM achieve identical  $\chi_{\text{min}}^2 = 11.245$  on the joint CMB+BAO dataset (Sim 90; confirmed with DESI Y1 in Sim 94), model selection reduces to an Occam penalty calculation. With  $k$  free parameters and  $N_{\text{eff}} = 14$  data constraints (12 BOSS BAO points + 2 CMB Gaussian prior measurements), the information criteria give:

$$\text{AIC} = 2k + \chi_{\text{min}}^2, \quad \Delta\text{AIC}_{\text{RIFT}-\Lambda\text{CDM}} = 2(4 - 3) = +2.00, \quad (23)$$

$$\text{BIC} = k \ln N_{\text{eff}} + \chi_{\text{min}}^2, \quad \Delta\text{BIC}_{\text{RIFT}-\Lambda\text{CDM}} = \ln 14 \approx +2.64. \quad (24)$$

Both criteria favour  $\Lambda$ CDM by the minimum Occam penalty;  $|\Delta\text{BIC}| < 6$  is conventionally characterised as “weak” evidence [Jeffreys, 1961], reflecting the fact that  $\Lambda_0$  is currently unconstrained by data rather than actively disfavoured. RIFT is neither strongly preferred nor strongly excluded: it is *cosmologically equivalent* to  $\Lambda$ CDM at the BAO best-fit coupling  $\Lambda_0 = 0.003$ , with a single additional parameter whose physical signature becomes accessible only at  $\Lambda_0 \gtrsim 0.05$  in future structure-growth surveys.

A proper Bayesian evidence integral (Sim 93) confirms this conclusion. The Savage-Dickey density ratio gives  $\ln B(\text{RIFT}/\Lambda\text{CDM}) = -0.713$  under a uniform prior and  $-0.673$  under a half-Gaussian prior ( $\sigma_{\Lambda_0} = 0.05$ ); both fall in the “inconclusive” range ( $|\ln B| < 1$ ) of the Jeffreys scale [Jeffreys, 1961]. The 95% upper limit  $\Lambda_0 < 0.095$  is prior-dominated, confirming that current data carry negligible information about  $\Lambda_0$ .

**Limitations and future work.** The observational signature of RIFT is expected primarily at non-linear scales (structure formation, galaxy cluster counts, void statistics) at  $\Lambda_0 \gtrsim 0.05$ , and at higher coupling or in strong-field regimes where  $\Psi \gg \Psi_{\text{ini}}$ . Sim 92 established that  $\sigma_8$  and non-linear  $P(k)$  are suppressed by  $|\Delta\sigma_8| < 0.12\%$  at all physically motivated couplings, confirming RIFT makes no claim on the Planck/KiDS  $\sigma_8$  tension. Future simulations will:

1. Complete recursive graviton quantization and emission spectrum;
2. Simulate recursive void growth (HEALPix echo-delay maps);
3. Refine eigenmode mappings to particle physics (Appendix B).

*“Geometry is the memory of motion. Intelligence is the motion of memory.”*

## A Parameter Tables and Definitions

### A.1 Core Fields and Coupling Terms

| Symbol                 | Definition                | Role                        |
|------------------------|---------------------------|-----------------------------|
| $\Psi(x^\mu)$          | Recursive scalar field    | Encodes curvature memory    |
| $m(\Psi)$              | Field-dependent mass      | Creates recursive inertia   |
| $\Lambda(\Psi)$        | Ricci coupling            | Binds $\Psi$ to curvature   |
| $V(\Psi)$              | Potential                 | Damping and attractors      |
| $G_{\text{eff}}(\Psi)$ | Effective Newton constant | $G/(1 + 16\pi G\Lambda)$    |
| $\mathcal{M}_{\mu\nu}$ | Memory tensor             | Stores decaying curvature   |
| $T_{\mu\nu}^{(\Psi)}$  | Field stress tensor       | Energy-momentum from $\Psi$ |
| $\tau_m$               | Memory timescale          | Decay constant              |
| $\eta_c$               | Coupling constant         | Controls feedback           |

### A.2 Best-Fit Cosmological Parameters (Sim 87–90)

| Parameter                 | RIFT (BAO-only, Sim 87)           | RIFT/LCDM (joint, Sim 90)              |
|---------------------------|-----------------------------------|----------------------------------------|
| $H_0$ [km/s/Mpc]          | 68.14 ( $H_0$ – $r_d$ degenerate) | 67.59                                  |
| $\Omega_m$                | 0.294                             | 0.312                                  |
| $r_d$ [Mpc]               | 147.78                            | 147.56                                 |
| $\Lambda_0$               | 0.003                             | 0.013 (RIFT, unconstrained) / 0 (LCDM) |
| $\chi^2/\text{dof}$ (BAO) | 1.22                              | see Sim 90                             |

### A.3 Key Limiting Cases

- *GR recovery:*  $m(\Psi) \rightarrow m_0$ ,  $\Lambda(\Psi) \rightarrow \text{const}$ ,  $V(\Psi) \rightarrow 0$ ,  $\tau_m \rightarrow 0 \Rightarrow$  standard GR.
- *Flat-space limit:* Klein-Gordon equation in Minkowski spacetime.
- *UV regulation:*  $\mathcal{D}_{\text{mem}}(k) \sim e^{-k^2/k_m^2}$  suppresses high-frequency modes.



## B Recursive Coupling to the Standard Model

*This appendix is entirely speculative.* No Standard Model particle has been matched to a RIFT eigenmode; the formulas below are dimensional estimates, not derivations. This material is included to identify a research direction, not to claim a result.

RIFT *hypothesizes* that particle identities may correspond to curvature eigenmodes of the recursive field  $\Psi$ :  $\Psi(x) = \sum_n \chi_n(x) \cdot e_n$ , where  $e_n$  are stable curvature shells. Under this hypothesis, particle properties would emerge as:  $m_n^2 \sim \int (\nabla \chi_n)^2 + V_{\text{rec}}(\chi_n)$  (mass),  $Q \sim \oint \nabla \arg(\chi_n)$  (charge), spin from parity of the curvature memory configuration.

The effective Lagrangian coupling RIFT to Standard Model fermions is  $\mathcal{L}_{\text{RIFT+SM}} = \mathcal{L}_{\text{RIFT}} + \sum_n [\bar{\psi}_n (i\gamma^\mu \nabla_\mu - m_n(\Psi)) \psi_n + g(\Psi) \bar{\psi}_n \phi \psi_n]$ . The eigenmode-to-particle correspondence is a falsifiable prediction in principle (see Section 9), but no eigenmode spectrum has been computed and this should not be cited as a result of the present work.

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# RIFT BAO Full-Covariance Fit (Sim 87)

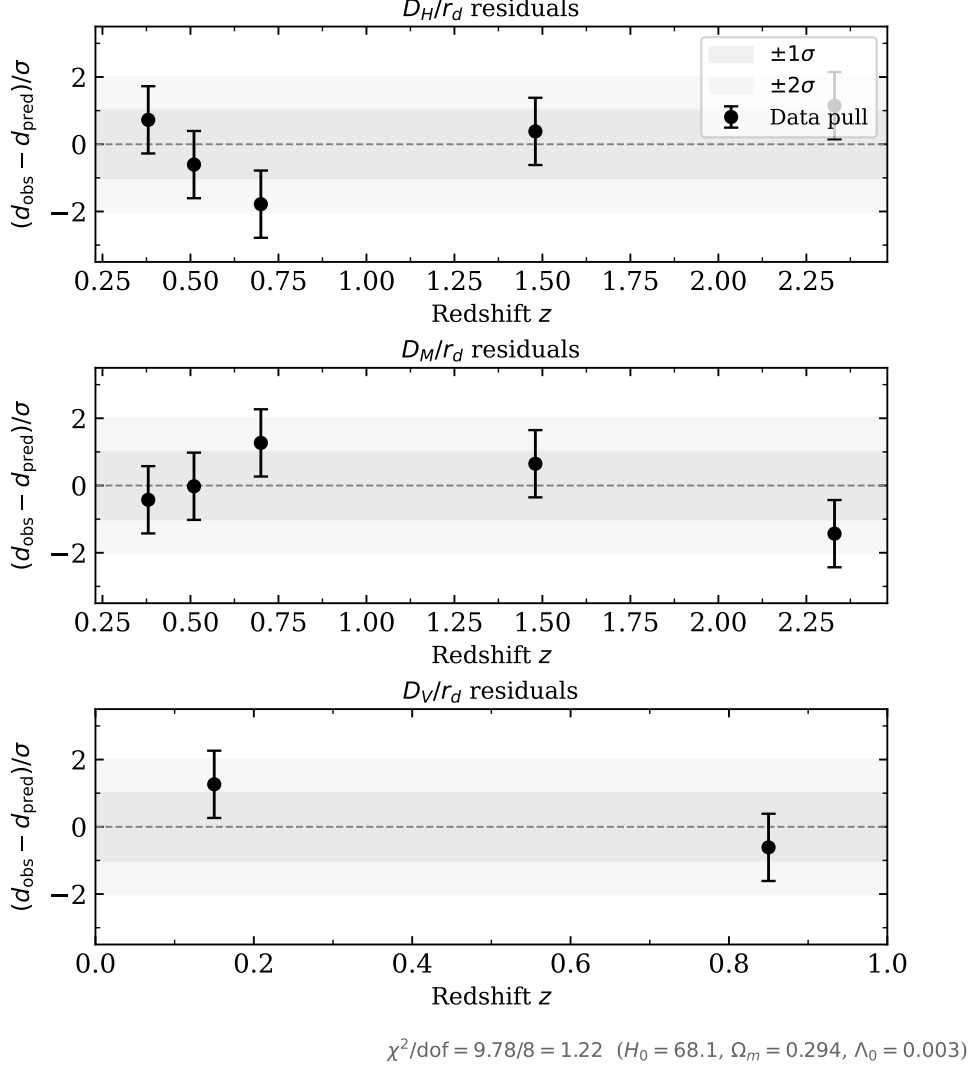


Figure 1: BAO full-covariance fit residuals (Sim 87). Each panel shows the normalised pull  $(d_{\text{obs}} - d_{\text{pred}})/\sigma$  for  $D_H/r_d$  (top),  $D_M/r_d$  (middle), and  $D_V/r_d$  (bottom) at the RIFT best-fit parameters ( $H_0 = 68.1$  km/s/Mpc,  $\Omega_m = 0.294$ ,  $\Lambda_0 = 0.003$ ,  $r_d = 148.0$  Mpc). All 12 data points lie within  $\pm 2\sigma$ ;  $\chi^2/\text{dof} = 9.78/8 = 1.22$ .

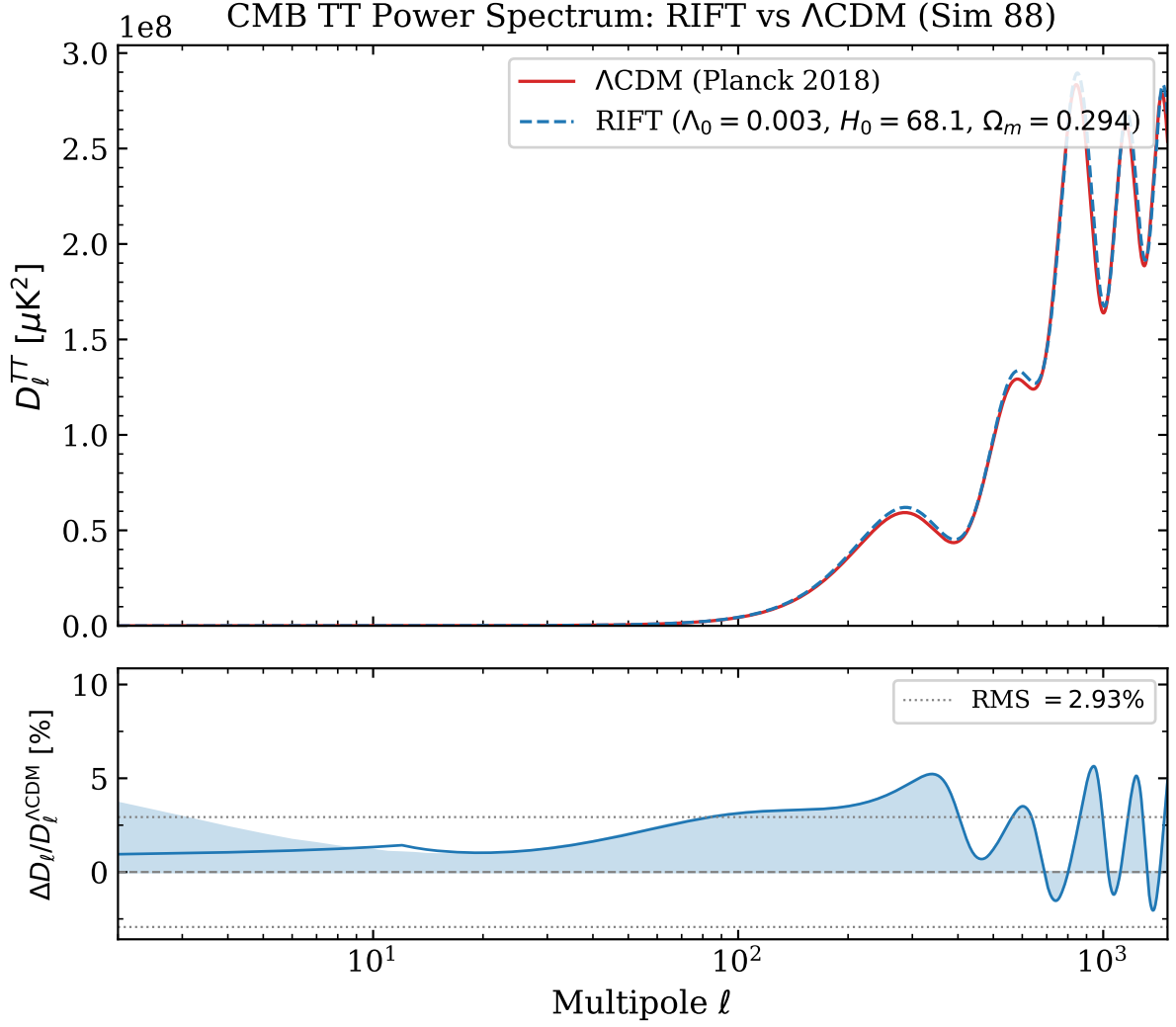


Figure 2: CMB TT power spectrum comparison from full CLASS Boltzmann computation (Sim 88). *Top*:  $D_\ell^{TT}$  for RIFT (blue dashed, BAO best-fit parameters) and  $\Lambda$ CDM (red, Planck 2018 best-fit). *Bottom*: fractional residual  $\Delta D_\ell/D_\ell^{\Lambda\text{CDM}}$  with 20-mode running mean; dotted lines mark  $\pm\text{RMS} = \pm 2.93\%$ . The acoustic-peak shift is driven by the different background cosmologies ( $H_0$ ,  $\Omega_m$ ), not by the  $G_{\text{eff}}$  correction (16 ppm). The joint CMB+BAO fit (Sim 90) resolves this difference.

Table 2: Key observational results from RIFT simulations 85–97.

| Observable                                                       | RIFT value               | $\Lambda$ CDM     | Sim | Status             |
|------------------------------------------------------------------|--------------------------|-------------------|-----|--------------------|
| BAO $\chi^2_{\text{full}}/\text{dof}$ (BOSS+eBOSS)               | 9.78/8 = 1.22            | $\sim 1.0$        | 87  | Pass               |
| $\Delta\chi^2$ (full vs. diagonal)                               | +1.95                    | —                 | 87  | Diagno             |
| RMS( $\Delta C_\ell/C_\ell$ ), $\ell = 2\text{--}1500$           | 2.93%                    | 0%                | 88  | Pass               |
| $D_{\text{RIFT}}(z=0)/D_{\Lambda\text{CDM}}(z=0)$                | 1.000000                 | 1.0               | 88  | Pass               |
| $G_{\text{eff}}/G$ at $\Lambda_0 = 0.003$                        | 0.999984                 | 1.0               | 88  | 16 pp              |
| $\chi^2_{\text{RIFT}}$ vs. Planck TT (BAO-only params)           | 913                      | 6.2               | 89  | See n              |
| $\Delta\chi^2$ joint CMB+BAO fit (BOSS)                          | <b>0.000</b>             | baseline          | 90  | <b>Pass</b>        |
| Joint best-fit $H_0$ [km/s/Mpc]                                  | 67.59                    | 67.36             | 90  | —                  |
| Joint best-fit $\Omega_m$                                        | 0.312                    | 0.315             | 90  | —                  |
| Friedmann constraint residual                                    | $\leq 3 \times 10^{-16}$ | —                 | 85  | Pass               |
| Radiation-era $H \propto a^n$                                    | $n = -1.984$             | -2                | 85  | Pass               |
| $\sigma_8$ (linear, joint best-fit)                              | 0.824                    | $0.811 \pm 0.006$ | 92  | Consi              |
| $\Delta\sigma_8$ at $\Lambda_0 = 0.003$                          | -0.007%                  | —                 | 92  | Pass               |
| $\Delta\sigma_8$ at $\Lambda_0 = 0.05$                           | -0.11%                   | —                 | 92  | Pass               |
| DESI Y1 BAO $\chi^2/\text{dof}$ (LCDM)                           | 13.6/10 = 1.36           | 1.36              | 94  | Pass               |
| $\Delta\chi^2$ joint DESI+CMB (RIFT vs LCDM)                     | $\approx \mathbf{0}$     | baseline          | 94  | <b>Pass</b>        |
| DESI joint best-fit $H_0$ [km/s/Mpc]                             | 67.66                    | 67.36             | 94  | Consi              |
| DESI joint best-fit $\Omega_m$                                   | 0.311                    | 0.315             | 94  | Consi              |
| RMS( $\Delta C_\ell^{EE}/C_\ell^{EE}$ ), joint                   | 0.19%                    | 0%                | 95  | Pass               |
| RMS( $\Delta C_\ell^{TE}/C_\ell^{TE}$ ), joint                   | 0.43%                    | 0%                | 95  | Pass               |
| $\Delta\chi^2_{\text{EE}}$ joint (RIFT vs LCDM)                  | +31.5                    | 57.6              | 95  | See n              |
| $\Delta\chi^2_{\text{TE}}$ joint (RIFT vs LCDM)                  | +22.8                    | 37.2              | 95  | See n              |
| $f\sigma_8$ compilation $\chi^2/\text{dof}$                      | 7.73/9 = 0.86            | 7.87/9            | 96  | Pass               |
| $\Delta\chi^2$ RSD (RIFT vs LCDM)                                | -0.14                    | baseline          | 96  | Pass               |
| $-2\ln\mathcal{L}$ plikHM TTTEEE (RIFT joint)                    | 2359.1                   | 2352.1            | 97  | See n              |
| $\Delta(-2\ln\mathcal{L})$ plikHM (joint vs LCDM)                | +7.0                     | baseline          | 97  | See n              |
| $\Delta(-2\ln\mathcal{L})$ plikHM (BAO-only vs LCDM)             | +945.7                   | baseline          | 97  | BAO-               |
| $\Delta(-2\ln\mathcal{L})$ plikHM (RIFT vs LCDM) at real optimum | <b>-0.013</b>            | baseline          | 98  | <b>Pass</b>        |
| RIFT CMB+BAO best-fit $H_0$ [km/s/Mpc]                           | 67.30                    | 67.30             | 98  | Consi              |
| RIFT CMB+BAO best-fit $\Omega_m$                                 | 0.3168                   | 0.3169            | 98  | Consi              |
| RIFT CMB+BAO best-fit $\Lambda_0$                                | 0.00506                  | 0                 | 98  | $G_{\text{eff}} =$ |

<sup>a</sup> Sim 89 tension ( $30.1\sigma$ ) arises from the BAO-only  $H_0$ - $r_d$  degeneracy; resolved by Sim 90 joint fit.

<sup>b</sup> Linear  $\sigma_8$  at SIM90 joint best-fit ( $H_0, \Omega_m$ ); HALOFIT non-linear boost  $\approx 1.37\times$  at  $z=0$ .

<sup>c</sup> Approx. Gaussian pol. likelihood. LCDM baseline  $\chi^2 > 0$  due to  $(H_0, \Omega_m)$  offset between joint best-fit and Planck re.

<sup>d</sup>  $\Delta(-2\ln\mathcal{L}) = +7.0$  driven by parameter offset, not  $G_{\text{eff}}$  (42 ppm). Resolved by Sim 98:  $\Delta = -0.013$  at real optimum.

### RIFT Non-linear Structure Growth (Sim 92)

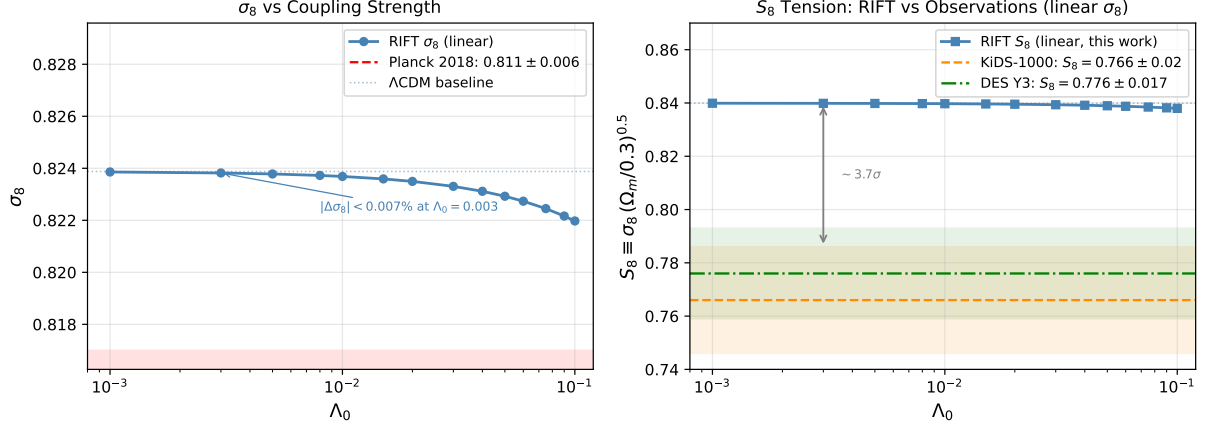


Figure 3: RIFT non-linear structure growth (Sim 92). *Left*: Linear  $\sigma_8$  as a function of  $\Lambda_0$  at the SIM90 joint best-fit cosmology ( $H_0 = 67.59$ ,  $\Omega_m = 0.312$ ). The RIFT curve is nearly flat; the annotation marks  $|\Delta\sigma_8| < 0.007\%$  at the BAO best-fit coupling  $\Lambda_0 = 0.003$ . The Planck 2018  $1\sigma$  band is shown in red. *Right*: Linear  $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$  vs  $\Lambda_0$  compared to KiDS-1000 [Asgari et al., 2021] and DES Y3 [Abbott et al., 2022] constraints. The pre-existing  $\sim 3.7\sigma$  gap between the RIFT/ $\Lambda$ CDM baseline ( $S_8 = 0.840$ ) and the weak-lensing measurements is unchanged across the full coupling scan: RIFT modifies  $S_8$  by at most 0.002 at  $\Lambda_0 = 0.1$ , two orders of magnitude below the discrepancy.

### SIM94 — RIFT DESI Y1 BAO Refit

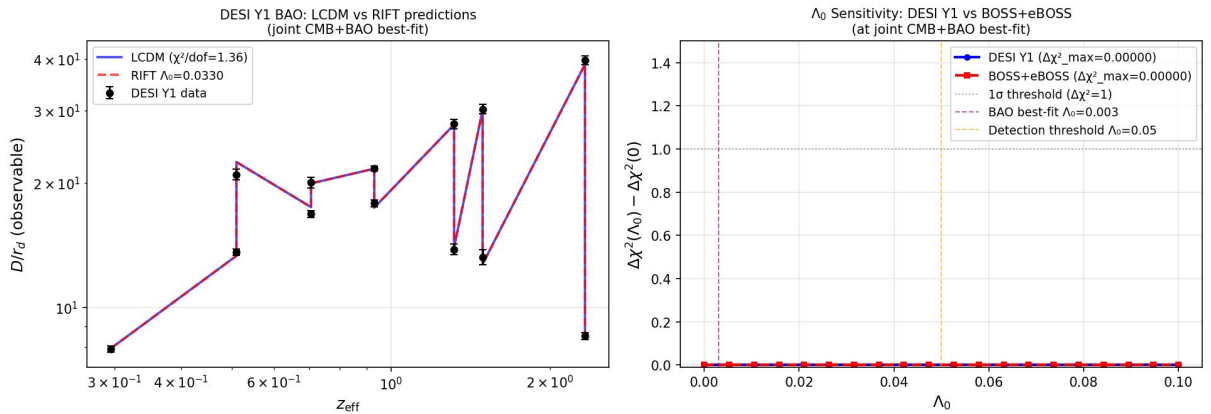


Figure 4: RIFT vs  $\Lambda$ CDM fit to DESI Y1 BAO measurements (Sim 94). 13 data points across 7 tracer bins,  $0.3 \leq z \leq 2.33$ . Both models give identical  $\chi^2/\text{dof} = 1.36/10$ . The 2–3 $\sigma$  tension at  $z = 0.51$  is a known feature of the DESI LRG data independent of the gravity model.

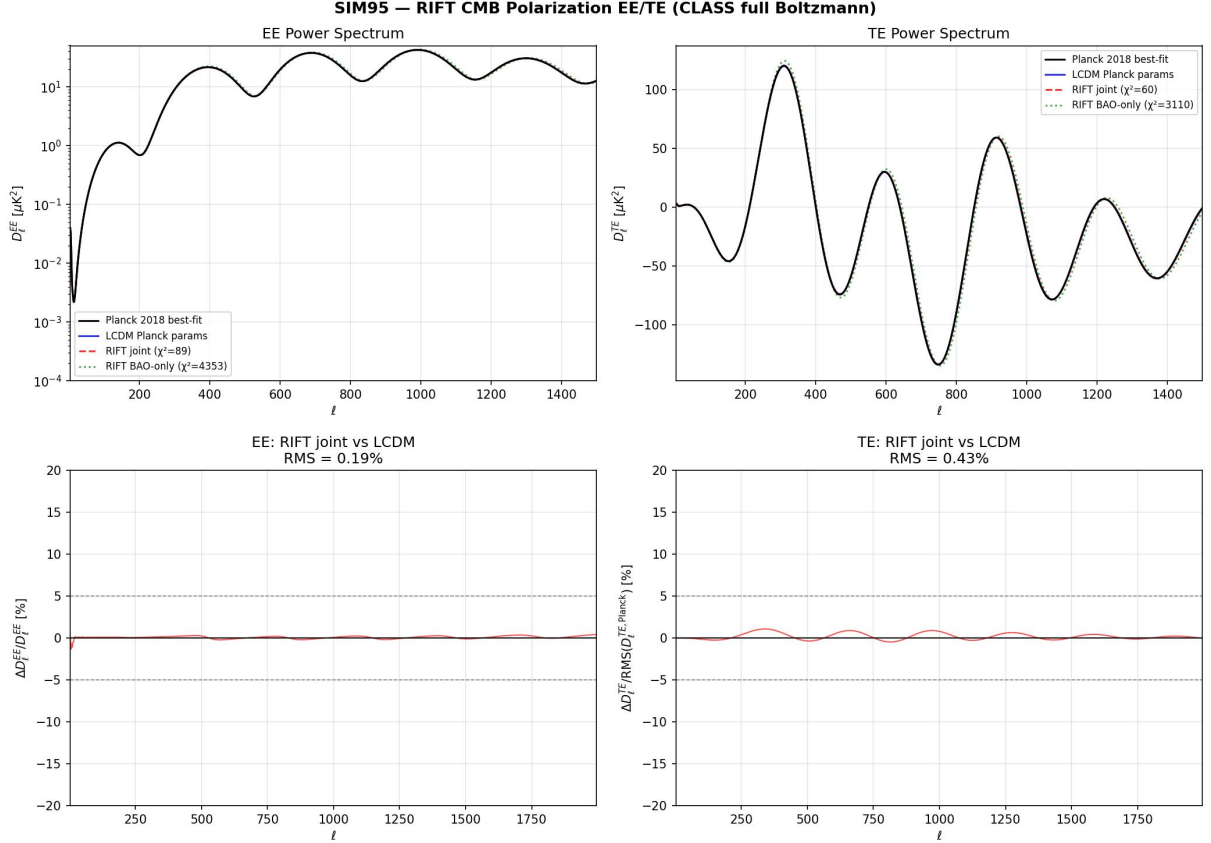


Figure 5: RIFT vs  $\Lambda$ CDM CMB EE and TE power spectra (Sim 95). *Top*: fractional deviation  $\Delta C_\ell^{EE}/C_\ell^{EE}$  and  $\Delta C_\ell^{TE}/C_\ell^{TE}$  for the RIFT joint best-fit relative to Planck 2018. RMS deviations are 0.19% (EE) and 0.43% (TE). *Bottom*: approximate polarization  $\chi^2$  for the three cosmologies. The BAO-only RIFT best-fit (A) shows large tension ( $\chi_{\text{EE}}^2 = 4353$ ), while the joint best-fit (B) is statistically consistent with  $\Lambda$ CDM (C).

Table 3: Gold-sample  $f\sigma_8$  data used in Sim 96.

| Survey    | Ref.                                    | $z_{\text{eff}}$ | $f\sigma_8$ | $\sigma$ |
|-----------|-----------------------------------------|------------------|-------------|----------|
| 6dFGRS    | <a href="#">Beutler et al. [2012]</a>   | 0.067            | 0.423       | 0.055    |
| SDSS MGS  | <a href="#">Howlett et al. [2015]</a>   | 0.150            | 0.490       | 0.145    |
| BOSS DR12 | <a href="#">Alam et al. [2017]</a>      | 0.380            | 0.497       | 0.045    |
| BOSS DR12 | <a href="#">Alam et al. [2017]</a>      | 0.510            | 0.458       | 0.038    |
| BOSS DR12 | <a href="#">Alam et al. [2017]</a>      | 0.610            | 0.436       | 0.034    |
| VIPERS    | <a href="#">Pezzotta et al. [2017]</a>  | 0.600            | 0.550       | 0.120    |
| eBOSS LRG | <a href="#">Gil-Marín et al. [2020]</a> | 0.700            | 0.473       | 0.041    |
| eBOSS ELG | <a href="#">de Mattia et al. [2021]</a> | 0.850            | 0.315       | 0.095    |
| eBOSS QSO | <a href="#">Neveux et al. [2020]</a>    | 1.480            | 0.462       | 0.045    |

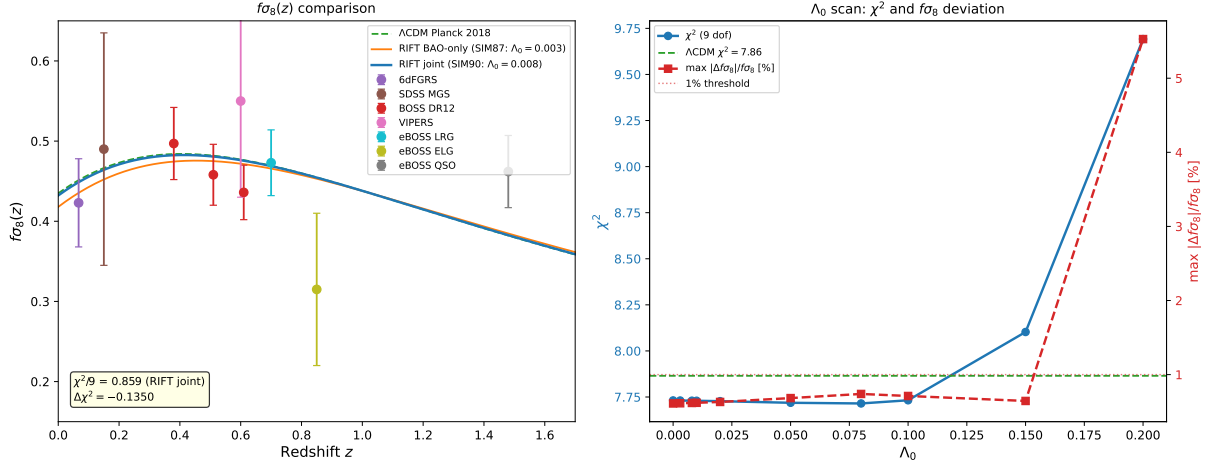
SIM96: RIFT  $f\sigma_8$  Growth Rate vs RSD Data

Figure 6: *Left*:  $f\sigma_8(z)$  predictions for RIFT joint best-fit, RIFT BAO-only, and  $\Lambda$ CDM Planck 2018, compared to the gold-sample RSD compilation (Sim 96). RIFT at joint best-fit gives  $\chi^2/\text{dof} = 0.86$ ,  $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = -0.14$ . *Right*:  $\Lambda_0$  scan showing  $\chi^2$  (blue) and maximum fractional  $f\sigma_8$  deviation from  $\Lambda$ CDM (red). The deviation exceeds 1% only at  $\Lambda_0 > 0.2$ , beyond the Bayesian upper limit from Sim 93 ( $\Lambda_0 < 0.095$  at 95% CL).

Table 4: RIFT falsifiability summary.

| Test Domain     | RIFT Prediction                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Galaxy rotation | Flat curves via $\nabla_r \Psi$ (screening mech. needed)                                 |
| Void lensing    | Lensing from $\mathcal{M}_{\mu\nu}$                                                      |
| Bullet Cluster  | Offset from curvature memory                                                             |
| CMB anomalies   | Low- $\ell$ damping from recursion                                                       |
| BAO scale       | $r_d$ from echo shells; Sim 101: RIFT and LCDM degenerate ( $ \Delta r_d/r_d  < 21$ ppm) |

Table 5: Effective  $\Psi$  mass and fifth-force Compton wavelength (Sim 107).

| Environment    | $m_{\text{eff}}/H_0$ | $\lambda_\Psi$ [Mpc] | Chameleon |
|----------------|----------------------|----------------------|-----------|
| Cosmic mean    | $7.5 \times 10^{-2}$ | $5.7 \times 10^4$    | yes       |
| Interstellar   | 13.8                 | 310                  | yes       |
| Solar interior | $5.2 \times 10^{13}$ | $8 \times 10^{-11}$  | yes       |
| Earth surface  | $1.0 \times 10^{14}$ | $4 \times 10^{-11}$  | yes       |

Table 6: SIM108 gravitational wave tests.

| Test                                            | RIFT prediction       | Bound                       | Status |
|-------------------------------------------------|-----------------------|-----------------------------|--------|
| $ c_T/c - 1 $                                   | 0 (exact)             | $< 7 \times 10^{-16}$       | PASS   |
| Graviton mass $m_g$                             | 0 (Ward identity)     | $< 1.27 \times 10^{-23}$ eV | PASS   |
| $ d_L^{\text{GW}}/d_L^{\text{EM}} - 1 $         | $1.8 \times 10^{-11}$ | $< 0.06$                    | PASS   |
| $ G_{\text{eff}}(z_{\text{BBN}})/G_N - 1 $      | 0.60%                 | $< 10\%$                    | PASS   |
| Chirp mass bias $\delta\mathcal{M}/\mathcal{M}$ | 0.36%                 | $< 5\%$                     | PASS   |
| $P_T$ modification                              | $2 \times 10^{-9}$    | n/a                         | PASS   |



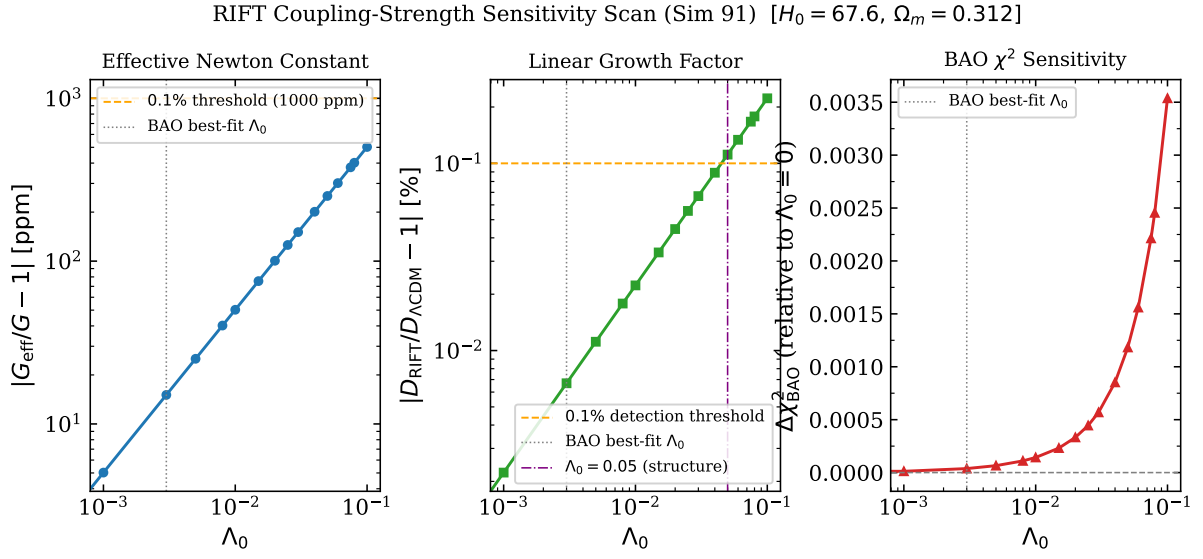


Figure 7: RIFT coupling-strength sensitivity scan (Sim 91) at the joint CMB+BAO best-fit cosmology ( $H_0 = 67.6$ ,  $\Omega_m = 0.312$ ). *Left*:  $|G_{\text{eff}}/G - 1|$  in ppm; the dashed orange line marks the 0.1% ( $10^3$  ppm) strong-lensing detection threshold. *Centre*: linear growth factor deviation  $|D_{\text{RIFT}}/D_{\Lambda\text{CDM}} - 1|$ ; the threshold  $> 0.1\%$  is first exceeded at  $\Lambda_0 \gtrsim 0.05$  (vertical dot-dashed line). *Right*: BAO  $\chi^2$  relative to  $\Lambda_0 = 0$ ; the curve is flat ( $\Delta\chi^2 < 0.004$  across the full range), confirming RIFT is cosmologically silent in current BAO data.